

# GUARDED WIN STATISTICS FOR CONTINUOUS AGENT EVALUATION

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## ABSTRACT

Agent evaluation combines verified completion, policy compliance, and resource use, often while trajectories are still arriving. A hierarchical win statistic makes outcome priorities explicit, but a favorable aggregate can conceal a decline in a higher-priority population outcome. We develop a guarded evaluation protocol that combines prioritized comparisons with separate component requirements. It distinguishes same-task benchmark repetitions from randomized cross-arrival online comparisons and gives finite-sample continuous-monitoring guarantees. For delayed outcomes, feasible completions of partial traces bound the final pair score. A prefix envelope transfers established full-data guarantees to these partial scores and cannot decide later than its matched completed-prefix comparator. Complete proofs identify the required enrollment-order assumptions and distinguish threshold error control from a calendar-time e-process. In 2,000-repetition stationary simulations, guarded betting made false deployments in 0.45% of identical-agent runs versus 31.05% for repeated ordinary Wald tests. In a separate informative-delay study, partial evidence reduced mean capped decision time from 41.285 to 22.363 ticks for an equally successful, cheaper alternative; both valid methods deployed in 99.7% of runs and had 0.6% null false deployments. An ordinal replay of 10,008 historical trace comparisons certified 34.2% before the later terminal marker, demonstrating archived certificate feasibility without claiming operational latency savings. Historical comparisons also expose success losses hidden by aggregate wins. These results support explicit decisions for a specified workload and preference rule, with unresolved outcomes retained in the analysis.

## 1 INTRODUCTION

An agent can finish a task while using excessive resources, violate a permission rule while obtaining a correct answer, or add verification steps that improve reliability at a higher cost. An evaluator therefore has several outcomes, often with a meaningful order of importance. Accuracy–cost analyses emphasize that capability alone does not characterize an agent system (Kapoor et al., 2024). Interactive agent benchmarks further expose variation across repeated trajectories and across task families (Barres et al., 2025; Yang et al., 2024). These measurements leave a consequential statistical question: when does accumulating evidence justify replacing the deployed system?

Generalized pairwise comparisons and win statistics address ordered outcomes in clinical trials (Buyse, 2010; Pocock et al., 2012). A comparison proceeds through a prespecified hierarchy, stopping at the first meaningful difference. Applied to agents, the rule might prefer policy compliance, then verified completion, then lower cost among successful runs. Unlike a weighted score, the explanation of a win identifies the deciding tier. This construction is established statistical machinery, and win statistics have already appeared in AI evaluation (Feng et al., 2026).

Three distinctions determine whether such a comparison can support an online decision. First, priority within a pair does not protect a population-level component: many cost wins can outweigh fewer success losses. Second, a resettable benchmark observes both systems on a common task, whereas an ordinary live A/B experiment observes only the assigned system on each arrival. The corresponding preferences need not agree. Third, repeatedly inspecting a fixed-sample confidence interval does not preserve its error rate. Repeated runs, adaptive allocation, and delayed completion add further ways for the nominal comparison to differ from the one actually estimated.

We specify a protocol that keeps these decisions separate and then combines them. The inferential ingredients are existing bounded-score confidence sequences and betting processes (Howard et al., 2021; Waudby-Smith & Ramdas, 2024). Our contribution is a design-specific treatment of prioritized agent evaluation: explicit targets for paired and online data; a guarded deployment rule with the appropriate stationary or drifting error guarantee; partial-trace score bounds that retain pending episodes; and reproducible experiments examining compensation, informative delay, and dependence. We give complete derivations, including conditions under which the guarantees do not apply. We do not propose a new generic confidence-sequence inequality or claim that sequential win statistics are new.

## 2 WHAT IS A WIN, AND WHOSE OUTCOMES ARE COMPARED?

Let  $Y$  record an entire episode through a fixed evaluation horizon: verified outcomes, selected trajectory summaries, and resources. A chronological tool call is not an independent statistical unit. For tier  $k$ , an antisymmetric rule  $d_k(y, y') \in \{-1, 0, 1\}$  reports a meaningful advantage, disadvantage, or operational tie. The hierarchical kernel is

$$h(y, y') = \sum_{k=1}^K d_k(y, y') \prod_{\ell < k} \mathbf{1}\{d_\ell(y, y') = 0\}. \quad (1)$$

Only the first decisive tier contributes. A resource comparison can be disabled unless both episodes satisfy a symmetric eligibility rule, such as both achieving verified completion. For positive costs, our empirical primary tolerance is  $|c_A - c_B| > 0.05 \max(c_A, c_B)$ . Equality is a tie. This threshold is an operational preference, not a significance test.

For a specified comparison law, let  $(p_W, p_L, p_T)$  denote win, loss, and tie probabilities. We monitor the bounded net benefit  $\theta = p_W - p_L = \mathbb{E}h$ . We also report  $WR = p_W/p_L$  when  $p_L > 0$ , and  $WO = (1 + \theta)/(1 - \theta)$  when defined. A win ratio of two does not imply twice the task-success probability. Tier-level win and loss contributions accompany the summary, together with the component means. Zero losses do not justify adding an arbitrary pseudocount.

**Three comparison targets.** For task description  $X$  and episode randomness  $\xi$ , a resettable benchmark can estimate

$$\theta_{\text{task}} = \mathbb{E}_X \mathbb{E}_{\xi_A, \xi_B} h\{Y^A(X, \xi_A), Y^B(X, \xi_B)\}, \quad \xi_A \perp \xi_B \mid X. \quad (2)$$

This compares independent runs of the same task and depends on the task distribution. Using the same simulator seed instead defines a coupled-run preference. Neither is automatically the individual live-arrival cross-world quantity  $\mathbb{E}h(Y^A, Y^B)$ , which requires a joint potential-outcome law unobserved under single-system exposure. The distinction between marginal, individual, and same-covariate preferences is established in Even & Josse (2025). Contrast functions averaged over the marginal potential-outcome laws, with inverse-probability and doubly robust estimation, are developed by Mao (2018); this distributional causal target is not new here.

For independent online arrivals in stratum  $s$ , the observable distributional comparison is

$$\theta_s = \int h(a, b) dF_{A,s}(a) dF_{B,s}(b). \quad (3)$$

This can mix tasks of different difficulty within a stratum. Stratification by an outcome-blind task family refines the question but does not make separate users the same user. Appendix A supplies an explicit coupling example with identical arm marginals and opposite individual preferences. The example illustrates an identification limitation; it is not evidence that the product-law target in Equation (2) is inherently unidentified.

**Pairwise priority and population protection.** Let  $B$  always succeed. Suppose  $A$  fails with probability  $\epsilon$ , and otherwise succeeds more cheaply. Then a success-first hierarchy gives  $\theta = 1 - 2\epsilon$  while  $\Delta_S = P(S^A = 1) - P(S^B = 1) = -\epsilon$ . Thus even a strongly favorable preference can violate a success noninferiority margin. We define the desired deployment region as

$$\mathcal{D} = \{\theta > 0, \Delta_j > -\delta_j \text{ for all } j = 1, \dots, J\}, \quad (4)$$

where  $g_j(Y) \in [0, 1]$  is an independently reported component and  $\Delta_j = \mathbb{E}g_j(Y^A) - \mathbb{E}g_j(Y^B)$ . The margins  $\delta_j$  require application-specific justification. A relative compliance requirement also does not certify an acceptable absolute event rate.

### 3 A RANDOMIZED ONLINE OBSERVATION PROTOCOL

Before observing outcomes, form disjoint pairs of arrivals within a prespecified stratum. Let  $R_i = 1$  assign system  $A$  to the first arrival and  $B$  to the second;  $R_i = 0$  reverses the order. The known probability  $q_i = \mathbb{P}(R_i = 1 \mid \mathcal{F}_{i-1})$  may depend on completed past data, but is bounded away from zero and one. Each arrival receives only one system. This design fixes one exposure per arm per pair; it is not an arbitrary traffic-allocation bandit.

Write  $U_i = h(Y_{i1}^A, Y_{i2}^B)$  and  $V_i = h(Y_{i1}^B, Y_{i2}^A)$  for the potential oriented pair scores. Only one is observed. The inverse-probability score

$$Z_i = \frac{R_i U_i}{2q_i} + \frac{(1 - R_i)V_i}{2(1 - q_i)} \quad (5)$$

has conditional expectation  $\mathbb{E}\{(U_i + V_i)/2 \mid \mathcal{F}_{i-1}\}$  under randomization independent of the potential records and no cross-arrival interference. With conditionally independent, exchangeable arrivals in a stratum, this reduces to Equation (3). Otherwise it remains a symmetric cross-arrival target, rather than silently imposing exchangeability. Component scores replace  $h$  by  $g_j(a) - g_j(b)$ .

For balanced order randomization,  $q_i = 1/2$  and  $Z_i \in \{-1, 0, 1\}$ . For unbalanced randomized order,  $|Z_i| \leq B_i = 1/\{2 \min(q_i, 1 - q_i)\}$ . Unweighted observed preferences can be biased when the two positions differ. The target, weighting, and range must therefore be specified together. The proof is a conditional randomization identity (Theorem 4), not an assumption that adaptive allocation makes potential outcomes observable.

**Endpoint maturity.** Freeze an episode horizon  $H$ . Prespecified per-episode limits define endpoint outcomes; exhaustion of a study-level monetary budget instead leaves unobserved endpoints requiring bounds. A baseline processes the completely matured prefix of enrolled pairs. This can incur latency when a slow earlier pair blocks later completed pairs. It preserves the order defining the target. Selecting whichever runs finish first can preferentially include fast successes and change the comparison law. A display at a random completed-prefix index is covered by an all-prefix guarantee; this is not a license to delete missing or slow episodes. The delayed-outcome result in Proposition 9 states the necessary filtration assumptions.

### 4 CONTINUOUS MONITORING AND GUARDED DECISIONS

**A finite-sample bound for drifting averages.** Let  $\mu_i = \mathbb{E}(Z_i \mid \mathcal{F}_{i-1})$  and  $\bar{\mu}_n = n^{-1} \sum_{i \leq n} \mu_i$ . If the predictable range of  $Z_i$  has width  $r_i$ , put  $V_n = \sum_{i \leq n} r_i^2/4$ . For a fixed  $\rho > 0$ , a normal-mixture confidence sequence is

$$\bar{Z}_n \pm \frac{1}{n} \sqrt{(V_n + \rho) \log \left( \frac{V_n + \rho}{\rho \alpha^2} \right)}. \quad (6)$$

Simultaneously over all  $n$ , this interval covers  $\bar{\mu}_n$  with probability at least  $1 - \alpha$ . For balanced ternary scores  $V_n = n$ . The proof integrates Hoeffding exponential supermartingales against a normal mixing law and applies Ville's inequality (Howard et al., 2021). Under drift the covered quantity is a running average of conditional effects; it does not describe the current or future deployment population. Range-only bounds can be conservative for rare-event components.

**A directional betting implementation for stationary gates.** For a bounded score  $X_i \in [-B, B]$ , gate threshold  $c > -B$ , and fixed bets  $0 < \lambda_k < 1/(B + c)$  with weights  $w_k \geq 0$  summing to one, define

$$E_n(c) = \sum_k w_k \prod_{i=1}^n \{1 + \lambda_k (X_i - c)\}. \quad (7)$$

Under the conditional null  $\mathbb{E}(X_i \mid \mathcal{F}_{i-1}) \leq c$ , this is a nonnegative supermartingale starting at one. Crossing  $1/\alpha$  therefore rejects the gate null with anytime error at most  $\alpha$  (Waudby-Smith & Ramdas, 2024). Our stationary simulations use 40 fixed geometric bets and ternary count sufficient statistics. Finite-grid validity does not imply power for arbitrarily small gaps: at least one bet must have positive expected log growth. This implementation is a standard betting baseline specialized to the deployment rule.

**One deployment claim versus simultaneous component claims.** Let  $c_0 = 0$  for preference and  $c_j = -\delta_j$  for the component gates. Under constant conditional means, accept deployment only after every gate has valid evidence against its null. If the deployment region fails, at least one fixed component null is true. A false deployment entails a crossing of that component’s level- $\alpha$  process. Consequently the probability of ever falsely deploying is at most  $\alpha$ , without dividing  $\alpha$  across the required conjuncts (Theorem 7). This intersection–union guarantee concerns the single deployment decision; it is not simultaneous  $1 - \alpha$  coverage for all reported effects.

Under arbitrary drift, the component violating the region can change over time. Then a fixed-null argument is insufficient. Use simultaneous component confidence sequences, for example at levels  $\alpha/(J + 1)$ , and deploy only when every lower bound for its running target exceeds its threshold (Theorem 8). For several candidates or repeated model revisions, further error allocation is necessary. Reusing the same data to choose the hierarchy, tolerances, or candidate invalidates a guarantee that assumes those choices frozen.

#### 4.1 USING INCOMPLETE TRAJECTORIES WITHOUT SELECTING FAST OUTCOMES

Let  $N(t)$  denote the number of enrolled pairs at calendar time  $t$ . For each gate, the observed partial traces define bounds  $\ell_i(t) \leq X_i \leq u_i(t)$  on the final fixed-horizon score, within the known score range, simultaneously for every pair and update. These bounds range over every feasible completion. For a success-then-cost hierarchy, a successful  $A$  episode at cost 1 already defeats a pending  $B$  episode that has irreversibly spent 2:  $B$  either fails or succeeds at higher cost. An unresolved higher safety tier would prevent that certificate. An absence of observed violations alone cannot certify final compliance.

Write  $b_\alpha(V_n)$  for the numerator of the radius in Equation (6). For every enrolled prefix  $n \leq N(t)$ ,

$$\left[ \frac{\sum_{i \leq n} \ell_i(t) - b_\alpha(V_n)}{n}, \frac{\sum_{i \leq n} u_i(t) + b_\alpha(V_n)}{n} \right] \quad (8)$$

covers the corresponding running conditional mean simultaneously over  $n$  and  $t$  with probability at least  $1 - \alpha$ . On the event that the complete-score confidence sequence covers every prefix, replacing its score sum by lower and upper bounds preserves coverage pathwise. No outcome-independent reveal-time assumption is needed. Its width is the complete-data width plus  $n^{-1} \sum_i (u_i - \ell_i)$ , making the cost of unresolved information explicit. If the number enrolled tends to infinity and that average width vanishes, stationary means with strictly positive gate margins are eventually resolved (Appendix B).

For stationary gates, positive betting factors similarly give

$$\underline{E}_{n,t}(c) = \sum_k w_k \prod_{i \leq n} \{1 + \lambda_k(\ell_i(t) - c)\} \leq E_n(c). \quad (9)$$

Thus a partial-score crossing at any time and prefix entails a crossing of the latent complete-score process, controlling its probability under the fixed conditional null. Retrospectively refining a pending factor uses new information about the original pair; it does not add a new observation. Under stationarity, deploy if one available prefix satisfies all gates. Evaluating every candidate prefix includes every completed prefix and therefore cannot decide later than a completed-prefix method using the same candidates, bets, and calendar looks. Using only the latest enrolled prefix lacks this dominance, because additional pending pairs can dilute its lower wealth.

The argument assumes a valid complete-score enrollment filtration and guaranteed completion bounds. Shared infrastructure, candidate adaptation, or fallible interim labels can violate those premises. We prove threshold-crossing control, not that the partial wealth is an e-process in the richer calendar-time filtration. Worst-case pending-outcome stopping and time-uniform filtration transfer have direct

Table 1: Deployment probability (%) over 2,000 synthetic repetitions. Win-only betting addresses composite preference; guarded methods require the same three gates. Repeated Wald is unadjusted; group Wald uses ten planned looks with Bonferroni allocation. Wald validity is asymptotic. The normal-mixture guarded baseline made no deployments in these settings.

|                       | Win only<br>betting | Guarded<br>betting | Repeated<br>Wald | Group<br>Wald | Fixed<br>Wald |
|-----------------------|---------------------|--------------------|------------------|---------------|---------------|
| Identical agents      | 0.55                | 0.45               | 31.05            | 1.65          | 4.30          |
| Efficiency gain       | 100.00              | 96.85              | 99.95            | 99.30         | 99.95         |
| Success regression    | 100.00              | 0.00               | 0.35             | 0.00          | 0.00          |
| Compliance regression | 100.00              | 0.00               | 2.05             | 0.00          | 0.00          |
| Joint gain            | 100.00              | 100.00             | 100.00           | 100.00        | 100.00        |
| Weak gain             | 5.50                | 4.90               | 58.35            | 13.30         | 28.40         |

precedents (Henzi & Ziegel, 2022b;a; Choe & Ramdas, 2024); our specialization uses hierarchy and component bounds in a deployment protocol. Under drift, each selected prefix retains its own target and requires simultaneous component coverage. A maximum of historical lower bounds is not a lower bound on the current drifting effect.

**Offline repeated runs.** With  $R$  common-seed trials for each task, the  $R^2$  cross-run scores are dependent. Under independent seed clusters, pairs with different seed indices estimate the independent-run target; the  $R$  same-seed pairs estimate a coupled target. Their equal-weight mixture assigns weight  $1/R$  to the latter. We average eligible cross-run scores within each task, weight tasks equally, and resample complete tasks for uncertainty. The effective independent units are tasks, or larger shared clusters when required. The appendix provides the unbiased task estimator, asymptotic variance, bounded-mean alternative, and coupling decomposition.

## 5 EXPERIMENTS

**Synthetic online streams.** We generated six stationary independent-arrival scenarios, each with 2,000 independent repetitions and at most 10,000 pairs (20,000 agent executions). Compliance and success are Bernoulli outcomes; cost is lognormal. The primary hierarchy is compliance, success, then cost among jointly compliant successful pairs. The guarded rule requires positive net benefit, a success difference above  $-0.03$ , and a compliance difference above  $-0.01$ . These illustrative margins were fixed before viewing the initial numerical results. Alpha is 0.05. Monitoring begins at 100 pairs and occurs every 50 pairs, additionally including the ten prespecified group looks. Analytic net-benefit targets are independently checked in the code.

Under identical agents, guarded betting made 9/2,000 deployments (0.45%; 95% Wilson interval 0.24–0.85%), while unadjusted repeated Wald inspection made 621/2,000 (31.05%; 29.06–33.11%). The valid guarded betting procedure deployed the cheaper, equally effective agent in 96.85% of repetitions. Win-only betting favored the cheaper agent in every success-regression and compliance-regression repetition; guarded betting made no deployments in either scenario (0/2,000; upper Wilson limit 0.192%). These are failures of a preference-only deployment rule, not false rejections of the composite-preference null, which is false in those scenarios.

The range-only normal-mixture rule made no deployments in any initial scenario. Its width at 10,000 balanced pairs is about 0.033, too large to resolve the 0.01 compliance margin when the compliance difference is near zero. This is a documented conservatism cost of that baseline, not evidence that anytime-valid inference generally lacks power. Directional betting uses the observed score distribution and is materially more useful here. We do not infer uniform power superiority over competitive empirical-Bernstein or sequential U-statistic methods.

**Targeted stress tests.** Extensions designed after the initial run placed each guardrail exactly on its null boundary. Guarded betting deployed in 0.75% of success-boundary and 0.45% of compliance-boundary repetitions; repeated Wald deployed in 32.40% and 30.45%. A separate repeated-run experiment with 40 task clusters and four runs per arm gave 95.25% coverage for a task-level  $t$  interval, versus 55.65% when all 16 within-task comparisons were incorrectly treated as independent. In a

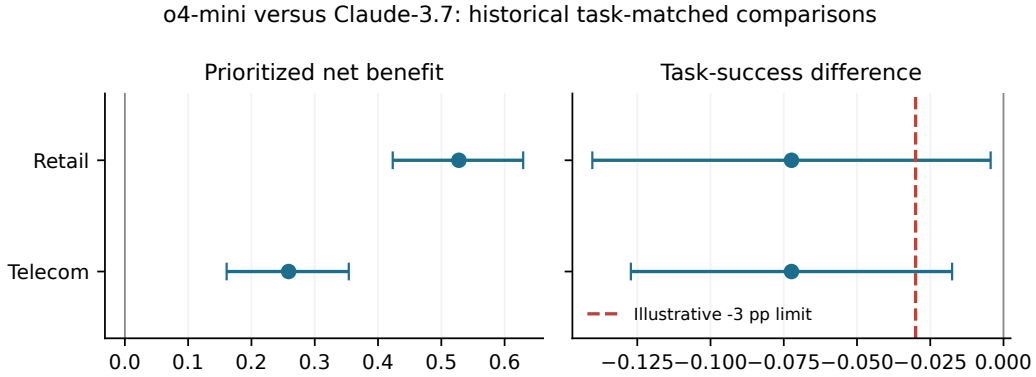


Figure 1: Historical agent comparisons: net benefit and success-rate difference can favor opposite systems. Intervals are pointwise task bootstrap summaries, conditional on the archived run configurations; they do not supply simultaneous or production-population guarantees.

nonexchangeable-pair stress model with adaptive order probabilities and zero symmetric preference, unweighted normal-mixture monitoring rejected in 93.0% of 1,000 repetitions. Correctly weighted betting rejected in 1.1% and the conservative weighted normal-mixture rule in 0%. This abstract bounded-score experiment isolates weighting bias; it does not simulate full agent traces.

**Historical agent trajectories.** We reanalyzed 3,336 public  $\tau^2$ -bench runs from three systems (GPT-4.1, o4-mini, and Claude-3.7) on 50 airline, 114 retail, and 114 telecom tasks, with four trials per task and system. Within each domain, the compared runs share the same harness commit, tasks, and user simulator. We also analyzed 600 SWE-agent trajectories, one per system on each of 300 SWE-bench Lite tasks. The hierarchy is verified success, historical recorded agent cost, then tool-call count; both failures tie. No policy-compliance label is fabricated where the source does not provide one. Recorded simulation duration includes simulator and service delays and is not used as clean agent latency.

The primary  $\tau^2$  statistic excludes same-seed diagonal comparisons, leaving 12 cross-seed comparisons per task. A theory audit motivated this amendment before the hierarchical numerical results were inspected. Same-seed and all-16-pair statistics are sensitivity analyses. We report pointwise percentile intervals from 10,000 complete-task bootstrap resamples. Because these are historical benchmark tasks with only four shared seed values per domain, the intervals are descriptive under task-resampling assumptions, conditional on the recorded seed suite; they do not account for arbitrary cross-task seed effects or establish uncertainty over newly generated seeds. SWE repository dependence poses an additional limitation. There is no multiplicity-adjusted leaderboard claim.

The primary empirical finding is a disagreement between preference and success. In retail, o4-mini versus Claude-3.7 has net benefit 0.528 (pointwise interval [0.423, 0.629]), despite a success difference of  $-0.072$  ([ $-0.140, -0.004$ ]). In telecom the corresponding values are 0.259 ([0.161, 0.354]) and  $-0.072$  ([ $-0.127, -0.018$ ]). The recorded success differences themselves violate the illustrative three-percentage-point requirement, even though the composite preferences are favorable. This is a descriptive diagnosis on the archived workload, not an anytime-valid deployment decision. Cost decides 60.0% of retail and 33.1% of telecom comparisons for this contrast.

The two reversals persist for cost tolerances from 0% to 20%. Changing the priority itself matters: placing tool-call count before cost changes the telecom o4-mini–Claude-3.7 net benefit from 0.259 to  $-0.126$ . This is a different preference question, not an estimator inconsistency. SWE-agent Claude-3-Opus versus GPT-4 has net benefit  $-0.113$  and success difference  $-0.063$  on the archived runs. These case studies retain historical model, harness, and price limitations; they do not assert a ranking of current models.

**Prospective workflow feasibility pilot.** We froze 12 telecom tasks and compared standard Haiku against the same model with a verification instruction, with a common Haiku user simulator, ran-

domized workflow order, deterministic task checks and a 40-step cap. The fixed monetary stopping rule produced 18/24 completed runs and 9/12 complete pairs; 12 completed runs reached the step cap. Each workflow succeeded on 2/9 completed tasks. Across all 12 planned tasks, conservative missing-outcome bounds are  $[-0.417, 0.083]$  for net preference and  $[-0.25, 0.25]$  for success difference (verification minus standard). These finite-cohort bounds are not sampling intervals. Accounted pilot cost was \$3.9456 within the frozen \$4 cap. The result demonstrates execution and measurement feasibility; low success and incomplete pairs preclude a superiority conclusion. Appendix H retains the full cohort, protocol amendment, costs and failure accounting.

## 5.1 INFORMATIVE DELAYS AND PARTIAL OUTCOMES

A separate frozen simulation isolates an asymmetric measurement pipeline. Both systems succeed with probability 0.90; their lognormal cost scales are equal under the null and have ratio 0.55 under the alternative. Failures and expensive traces from  $A$  are reported late, even when the systems have identical outcome distributions. We enroll 100 pairs per tick, up to 6,000, and reveal all outcomes within 20 ticks of enrollment. This directly simulates the balanced oriented score law, without running agents or assigning production users. Each of 1,000 repetitions per scenario is shared by three methods: completed-only selection, completed prefixes, and partial-score prefix inference. The two valid methods use the same candidate prefixes, positive bets, and same-prefix conjunction of preference superiority and success noninferiority at margin 0.03. Missing bits admit both outcomes; unrevealed cost signs admit all signs. The bounds do not infer hidden values from the informative reveal clock.

Under identical outcomes, completed-only selection falsely deployed in 1000 of 1,000 repetitions, whereas each valid method deployed in 6 (0.6%; Wilson 95% interval, 0.275–1.303%). Under the cheaper alternative, both valid methods deployed in 997 of 1,000 repetitions. Partial evidence reduced mean capped decision time from 41.285 to 22.363 ticks; the paired gain was 18.922 (Monte Carlo standard error 0.033). Mean initiated executions fell from 8130.8 to 4459.6. No partial decision occurred later than its comparator, and all 997 deployment paths were strictly earlier. Identical final decisions after complete revelation are guaranteed by the matched prefix construction, not evidence of universally high power. The magnitude of the gain depends on this particular reveal mechanism.

**Certificates from actual archived prefixes.** We replayed all nine retained  $\tau^2$  source files, comprising 3,336 episodes and 51,247 assistant messages. With final labels hidden until a terminal marker and pending resources bounded only by accrued values, 3,426 of 10,008 off-diagonal comparisons (34.2%) had a guaranteed sign before the later terminal marker. Of all comparisons, 25.9% resolved while at least one actual assistant message remained unseen; the rest of the early cases preceded only the artificial terminal marker. Every one of 195,171 prefix intervals contained the archived final sign. The replay uses ordinal message ticks, not observed concurrent time, and does not test a sequential deployment rule or estimate execution savings. Appendix F reports every contrast and the completion-set audit.

**Decision objectives and grader errors.** A separate 500-repetition ablation on each of six scenarios compares success-only, component-only, guarded-win, and fixed weighted-utility rules. These test different objectives. With identical agents, component-only noninferiority qualifies 78.4% of streams while guarded win qualifies 0.2%: satisfying component margins does not establish a preference improvement. The fixed utility  $0.8S + 0.1C + 0.1/(1 + \text{cost})$  approves the compliance-regression scenario in 32.6% of runs, correctly reflecting its positive utility target; adding component requirements reduces this to zero. No rule is uniformly more powerful across different objectives. Appendix G explains the objectives; the supplementary CSV tables retain all comparisons and population Pareto descriptions.

Four additional cases perturb success grading. With 5% false-positive errors for  $A$  and 5% false-negative errors for  $B$ , a true success contrast of  $-0.04$  becomes  $+0.012$  measured. Measured-label guarded inference approves 489/500 streams (97.8%), while true-label inference approves none. A conservative rule using the specified error bounds also approves none. Conversely, adverse differential grading prevents detection of a genuine efficiency gain. Optional-stopping validity for measured outcomes cannot repair a biased definition of success. The error bounds are known simulation inputs, not empirically validated grader guarantees.

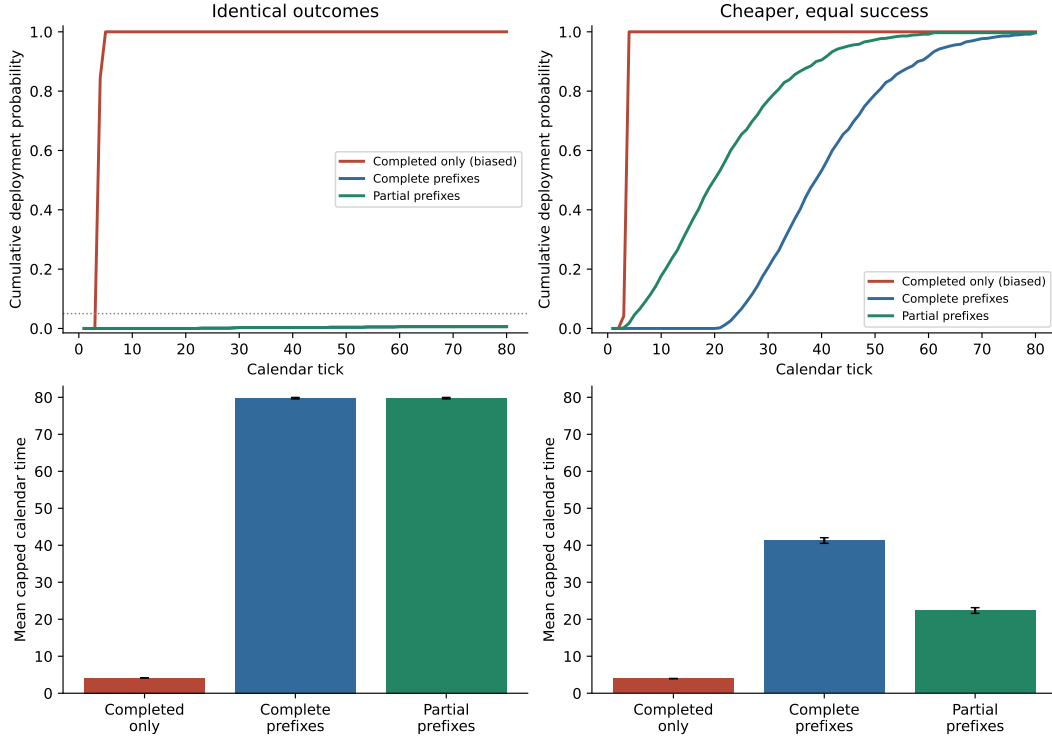


Figure 2: Informative-delay score simulation. Partial-score prefix inference retains the complete-data threshold guarantee while using incomplete episodes. Completed-only selection targets a biased subset. Calendar ticks and cost units are synthetic. Curves show the fraction that has deployed, using the same latent draws for all methods.

**An established likelihood-mixture reference.** A separate eight-scenario comparison uses the Dirichlet-mixture construction for categorical data (Lindon & Malek, 2022), with a fixed uniform prior and separate tests of the three raw gate scores. All methods share complete iid pairs, targets, thresholds and looks. With a fresh fixed seed and 2,000 repetitions, guarded DM has zero null deployments and 60.1% efficiency-gain deployments, versus 0.6% and 96.6% for guarded betting. Both detect the joint gain in every repetition. These are results of a separately reproduced comparison, not replacements for the primary study. Appendix I gives all eight scenarios, the likelihood derivation, and the narrower iid categorical assumptions. This reference is not substituted into partial-score betting, since its evidence is not coordinatewise monotone in the observed scores.

## 6 RELATED WORK AND SCOPE

Generalized pairwise comparisons (Buyse, 2010) and the clinical win ratio (Pocock et al., 2012) establish the ordered-outcome construction. Clustered win-statistic inference is also developed (Fang et al., 2026); benchmark tasks with both systems observed are not parallel-arm randomized clinical clusters. Sequential derived-win designs (Zhang & Wu, 2024; Bergemann & Hanson, 2026) and recent anytime-valid U-statistic methods (Cai et al., 2026) are direct methodological precedents. Our disjoint-pair construction trades some information available to all-pairs estimators for a simple randomized score and explicit finite-sample monitoring. We do not establish its efficiency relative to every sequential U-statistic method.

Two recent win-statistic proposals address nearby concerns. Li et al. (2026) tests benefit under every prespecified hierarchy or threshold rule using an intersection-union argument and asymptotic fixed-sample inference. Our conjunction uses separate outcome-component requirements and continuously monitored evidence; the intersection-union principle itself is established. McCoy et al. (2026) replaces reach-weighted tier aggregation with prespecified weights on stage-conditional effects, explicitly limiting lower-tier influence. That changes the composite estimand; it does not impose our

requirements on raw population success or compliance differences. A positive reweighted composite can still fail such a requirement, as illustrated in Appendix J. We claim neither discovery of lower-tier dominance nor superiority to these different decision targets.

Anytime off-policy confidence sequences already support deployment gates (Karampatziakis et al., 2021). Our use of those ideas is specific to prioritized episode comparisons and component constraints. The agent-evaluation contribution must therefore be judged by the clarity of the design, the empirical consequences it reveals, and the usefulness of the complete protocol, rather than novelty of e-processes themselves. Delayed-outcome inference also depends on the target: Lindon & Kallus (2026) studies calendar-time cumulative rewards, whereas our bounds retain fixed-horizon enrollment-prefix outcomes. Historical benchmark outcomes remain measurements under their original harness and model configurations. A favorable comparison does not identify the module causing a gain or establish transport to production.

**Limitations.** The primary synthetic model deliberately isolates compensation and monitoring; real workflows have richer dependence, measurement error, and workload shifts. Benchmark cost records are historical charges, not current price quotes. An imperfect success or policy grader changes the measured target, and rare severe failures require separate monitoring. Our component margins are illustrative, not elicited from application owners. All-pairs methods may use data more efficiently than disjoint pairs. Partial bounds can remain uninformative when outcomes cannot be certified early; our delay simulation uses one deliberately simple measurement pipeline and two outcome scenarios. The historical prefix audit validates certificate feasibility under an artificial message schedule; its operational latency and deployment value remain untested. Public retrospective traces and laboratory experiments do not constitute a randomized intervention among production users.

## REPRODUCIBILITY STATEMENT

The supplementary code contains the comparator, monitoring procedures, simulation generators, analytic simulation targets, seed and source manifests, and result tables. Appendix A states the assumptions and complete proofs. Public-data analysis records original artifact locations, hashes, processing rules, and task-level uncertainty. Internal protocol amendments are distinguished from public preregistration.

## ETHICS STATEMENT

This work studies synthetic environments and public benchmark records. Deployment rules require externally justified preferences and safeguards; a favorable composite cannot certify absolute safety or fairness. Released materials exclude credentials and private user data. No live production users are assigned experimental agents in the reported historical reanalyses or synthetic studies.

## AI USE STATEMENT

AI assistants were used substantially in literature discovery, conceptual development, theorem and proof drafting, experiment design and implementation, interpretation, manuscript drafting, and internal scientific review. These reviews are model-assisted checks, not independent human peer review. The present development version does not assert that human verification of every result and citation has been completed.

## REFERENCES

- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan.  $\tau^2$ -bench: Evaluating conversational agents in a dual-control environment, 2025. URL <https://arxiv.org/abs/2506.07982>. arXiv preprint.
- Tracy Bergemann and Tim Hanson. Group sequential methods for the win ratio, 2026. URL <https://arxiv.org/abs/2601.22525>. arXiv preprint.

- Marc Buyse. Generalized pairwise comparisons of prioritized outcomes in the two-sample problem. *Statistics in Medicine*, 29(30):3245–3257, 2010. doi: 10.1002/sim.3923. URL <https://doi.org/10.1002/sim.3923>.
- Leheng Cai, Qirui Hu, and Weijia Li. Asymptotic anytime-valid inference for U-statistics, 2026. URL <https://arxiv.org/abs/2605.14692>. arXiv preprint.
- Yo Joong Choe and Aaditya Ramdas. Combining evidence across filtrations, 2024. URL <https://arxiv.org/abs/2402.09698>.
- Mathieu Even and Julie Josse. Rethinking the win ratio: A causal framework for hierarchical outcome analysis, 2025. URL <https://arxiv.org/abs/2501.16933v4>. arXiv preprint; version 4 revised March 23, 2026.
- Xi Fang, Guangyu Tong, Yuan Huang, F. Perry Wilson, Patrick J. Heagerty, and Fan Li. Statistical inference with win statistics in cluster-randomized trials with hierarchical composite outcomes, 2026. URL <https://arxiv.org/abs/2604.18341v2>. arXiv preprint; version 2 revised September 16, 2026.
- Jean Feng, Vishal Patel, Patrick Heagerty, Yifan Mai, Venkatesh Sivaraman, Patrick Vossler, Jialin Ouyang, and Anupam B. Jena. Expert evaluation of clinical AI tools on real point-of-care clinical queries, 2026. URL <https://arxiv.org/abs/2606.28960>. arXiv preprint.
- Alexander Henzi and Johanna F. Ziegel. Correction to: ‘valid sequential inference on probability forecast performance’. *Biometrika*, 109(4):1181–1182, 2022a. doi: 10.1093/biomet/asac043.
- Alexander Henzi and Johanna F. Ziegel. Valid sequential inference on probability forecast performance. *Biometrika*, 109(3):647–663, 2022b. doi: 10.1093/biomet/asab047.
- Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991. URL <https://arxiv.org/abs/1810.08240>.
- Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. AI agents that matter, 2024. URL <https://arxiv.org/abs/2407.01502>. Citing the verified 2024 arXiv version.
- Nikos Karampatziakis, Paul Mineiro, and Aaditya Ramdas. Off-policy confidence sequences. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 5301–5310. PMLR, 2021. URL <https://proceedings.mlr.press/v139/karampatziakis21a.html>.
- Kexuan Li, Xue Fan, and Lingli Yang. Worst-case win ratios under partially specified outcome hierarchies, 2026. URL <https://arxiv.org/abs/2608.29857v1>. Version 1, August 30, 2026.
- Michael Lindon and Nathan Kallus. Design-based anytime-valid inference for randomized experiments with delayed outcomes and staggered entry, 2026. URL <https://arxiv.org/abs/2603.25971>. Version 2, May 29, 2026.
- Michael Lindon and Alan Malek. Anytime-valid inference for multinomial count data. In *Advances in Neural Information Processing Systems*, volume 35, pp. 2817–2831, 2022. URL [https://proceedings.neurips.cc/paper\\_files/paper/2022/hash/12f3bd5d2b7d93eadc1bf508a0872dc2-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2022/hash/12f3bd5d2b7d93eadc1bf508a0872dc2-Abstract-Conference.html).
- Lu Mao. On causal estimation using U-statistics. *Biometrika*, 105(1):215–220, 2018. doi: 10.1093/biomet/asx071. URL <https://doi.org/10.1093/biomet/asx071>.
- David McCoy, John Leopold, Shirley Galbiati, Minhthien Vu, and Bonnie Zhang. Priority-standardized net benefit: A stage-normalized estimand for hierarchical composite endpoints, 2026. URL <https://arxiv.org/abs/2607.22950v1>. Version 1, July 24, 2026.
- Stuart J. Pocock, Cono A. Ariti, Timothy J. Collier, and Duolao Wang. The win ratio: a new approach to the analysis of composite endpoints in clinical trials based on clinical priorities. *European Heart Journal*, 33(2):176–182, 2012. doi: 10.1093/eurheartj/ehr352. URL <https://doi.org/10.1093/eurheartj/ehr352>.

Aleksandr Podkopaev and Aaditya Ramdas. Sequential predictive two-sample and independence testing. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL [https://proceedings.neurips.cc/paper\\_files/paper/2023/hash/a72b207734d6112f6b47447e46be40e9-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2023/hash/a72b207734d6112f6b47447e46be40e9-Abstract-Conference.html).

Shubhanshu Shekhar and Aaditya Ramdas. Nonparametric two-sample testing by betting. *IEEE Transactions on Information Theory*, 70(2):1178–1203, 2024. doi: 10.1109/TIT.2023.3305867. URL <https://arxiv.org/abs/2112.09162v6>. See also the 2025 correction and corrected arXiv version 6.

Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 86(1):1–27, 2024. doi: 10.1093/jrssb/qkad009. URL <https://academic.oup.com/jrssb/article/86/1/1/7043257>.

John Yang, Carlos E. Jimenez, Alexander Wettig, Kilian Lieret, Shunyu Yao, Karthik Narasimhan, and Ofir Press. SWE-agent: Agent-computer interfaces enable automated software engineering. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/hash/5a7c947568c1b1328ccc5230172e1e7c-Abstract-Conference.html](https://proceedings.neurips.cc/paper_files/paper/2024/hash/5a7c947568c1b1328ccc5230172e1e7c-Abstract-Conference.html).

Baoshan Zhang and Yuan Wu. Sequential design with derived win statistics, 2024. URL <https://arxiv.org/abs/2410.06281>. arXiv preprint.

## A PROTOCOL, ESTIMANDS, AND COMPLETE PROOFS

This appendix gives complete proofs for the proposed protocol. The concentration and betting arguments are specializations of established anytime-valid inference, rather than new concentration inequalities (Howard et al., 2021; Waudby-Smith & Ramdas, 2024). Sequential win-statistic methods already exist (Zhang & Wu, 2024; Bergemann & Hanson, 2026), as do asymptotic anytime-valid methods for U-statistics (Cai et al., 2026) and confidence-sequence-based deployment gates (Karampatziakis et al., 2021). The purpose of the results below is to make the evaluation design, decision claim, and its limitations explicit.

### A.1 A MEASURABLE HIERARCHY WITH OPERATIONAL TIES

Let  $Y \in \mathcal{Y}$  denote the complete outcome record of an episode under a frozen evaluation protocol. It includes final outcomes, selected trajectory summaries, and resources measured through a fixed horizon. Let  $d_k : \mathcal{Y}^2 \rightarrow \{-1, 0, 1\}$ ,  $k = 1, \dots, K$ , be measurable, antisymmetric tier comparators:  $d_k(y, y') = -d_k(y', y)$ . A value of zero means a prespecified operational tie. For example, for a larger-is-better scalar endpoint  $g_k$  and an absolute tolerance  $\eta_k \geq 0$ ,

$$d_k(y, y') = \mathbf{1}\{g_k(y) - g_k(y') > \eta_k\} - \mathbf{1}\{g_k(y') - g_k(y) > \eta_k\}.$$

Resource tiers can be set to zero unless a symmetric eligibility condition is satisfied, such as both episodes achieving verified success. Such a rule is part of the estimand. It must not be chosen after inspecting the systems’ comparative results. Define

$$h_k(y, y') = d_k(y, y') \prod_{\ell < k} \mathbf{1}\{d_\ell(y, y') = 0\}, \quad h(y, y') = \sum_{k=1}^K h_k(y, y'). \quad (10)$$

At most one term is nonzero; hence  $h \in \{-1, 0, 1\}$  and  $h$  is antisymmetric. The first decisive tier determines the sign. A tolerance does not assert statistical equivalence, and absence of a significant difference is not used to define an episode-level tie.

For any specified joint comparison law  $P$  on  $(Y^A, Y^B)$ , put

$$p_W = P(h = 1), \quad p_L = P(h = -1), \quad p_T = P(h = 0), \quad \theta = E_P h = p_W - p_L.$$

The net benefit  $\theta$  is the primary bounded target. Tier contributions are  $\theta_k = E_P h_k$ , so that  $\theta = \sum_k \theta_k$ . The corresponding decisive masses  $P(|h_k| = 1)$  show how often each tier determined a comparison; these are not population-level component effects.

**Proposition 1** (Relations among win summaries). *Let  $d = p_W + p_L$ . When denominators are positive,*

$$\text{WR} = \frac{d + \theta}{d - \theta}, \quad \text{WO} = \frac{p_W + p_T/2}{p_L + p_T/2} = \frac{1 + \theta}{1 - \theta}.$$

*In particular,  $\text{WR} > 1$  is equivalent to  $\theta > 0$  when  $p_L > 0$ . The net benefit alone determines win odds, but does not determine the win ratio.*

*Proof.* Adding and subtracting  $d = p_W + p_L$  and  $\theta = p_W - p_L$  gives  $p_W = (d + \theta)/2$  and  $p_L = (d - \theta)/2$ . Also,  $p_W + p_T/2 = (1 + \theta)/2$  and  $p_L + p_T/2 = (1 - \theta)/2$ . The identities and equivalence follow. Different  $d \in [|\theta|, 1]$  give different win ratios at a fixed nonzero  $\theta$ .  $\square$

## A.2 PAIRWISE PRIORITY DOES NOT IMPLY A POPULATION GUARDRAIL

**Proposition 2** (Arbitrarily favorable preference can hide harm). *Consider verified success followed by cost, with lower cost preferred only when both systems succeed. For every  $\epsilon \in (0, 1)$  there is a comparison distribution with success-rate difference  $\Delta_S = P(S^A = 1) - P(S^B = 1) = -\epsilon$  and net benefit  $\theta = 1 - 2\epsilon$ . Consequently, for any noninferiority margin  $\delta \in [0, 1/2)$ , positive net benefit need not imply  $\Delta_S > -\delta$ . Moreover, net benefit can approach one while the success-rate difference remains strictly negative.*

*Proof.* Let  $B$  always succeed. Let  $A$  fail with probability  $\epsilon$  and otherwise succeed with lower cost than  $B$ . On failure,  $A$  loses at the success tier. On success,  $A$  wins at the cost tier. Thus  $p_W = 1 - \epsilon$ ,  $p_L = \epsilon$ , and  $\Delta_S = -\epsilon$ . For the margin claim choose  $\delta < \epsilon < 1/2$ . Taking  $\epsilon \downarrow 0$  proves the last claim. This construction works for same-task or independent-draw comparisons because  $B$ ’s outcome is constant.  $\square$

The proposition does not say that cost should be ignored. It shows that “success takes priority within a pair” and “population success cannot decline beyond a margin” are different requirements. We therefore use component scores  $g_j : \mathcal{Y} \rightarrow [0, 1]$ , oriented so larger values are better, and separate component differences  $\Delta_j$ . A safety score can be one minus an adverse-event indicator. An adverse-event rate that is already unacceptable for the reference also requires an *absolute* rate gate: noninferiority alone does not certify an acceptable absolute rate. The bounded-score results below also apply to an absolute score for  $A$ , by replacing a difference kernel with  $g_j(y^A)$ .

## A.3 WHAT AN ONLINE A/B EXPERIMENT IDENTIFIES

A live-episode counterfactual, a same-task clone comparison, and a two-arrival online comparison are three distinct targets. Let  $Y^A, Y^B$  first denote the potential outcomes of a single live episode and define  $\theta_{\text{episode}} = E h(Y^A, Y^B)$ . This target depends on the joint potential-outcome law, including latent episode features. For the clone design, let  $Y^a(x, \xi)$  be a complete episode under system  $a$ , task context  $x$ , and evaluation randomness  $\xi$ . The offline independent-replicate target is

$$\theta_{\text{task}} = E_X E_{\xi_A, \xi_B} h\{Y^A(X, \xi_A), Y^B(X, \xi_B)\}, \quad (11)$$

where the two replicate streams are conditionally independent. A common-randomness coupling defines a different target unless equality of these expectations is established. The law of  $X$ , evaluation horizon, and replication convention must all be reported.

For a stratum  $s$ , an online independent-draw target is

$$\theta_{\text{cross}, s} = E h(Y_s^A, \tilde{Y}_s^B), \quad Y_s^A \perp \tilde{Y}_s^B, \quad (12)$$

where each episode is drawn from that stratum’s workload. Even exact knowledge of both marginal outcome distributions generally does not identify  $\theta_{\text{episode}}$ . By contrast, if the arm-specific conditional distributions given an observed task context  $X$  are identified under overlap, their product defines the identifiable functional in (11). Individual-counterfactual, same-covariate independent-copy, and population independent-copy win targets, including the first target’s nonidentifiability, are explicitly distinguished by Even & Josse (2025). Our construction below illustrates that established obstruction for an agent episode.

**Proposition 3** (Episode-counterfactual preference is not identified). *There exist two full-data laws with identical distributions of all observed data from an independently randomized, one-system-per-task A/B experiment, but opposite  $\theta_{\text{episode}}$ . Both laws have independent-draw net benefit zero.*

*Proof.* Let the observed context  $X$  be constant, let the latent episode feature  $U$  be uniform on  $\{0, 1, 2\}$ , and use the single-tier comparator  $h(a, b) = \text{sign}(a - b)$ . Under the first law set  $Y^B = U$  and  $Y^A = (U + 1) \bmod 3$ . Under the second set  $Y^B = U$  and  $Y^A = (U - 1) \bmod 3$ . Under either law, each system’s marginal outcome is uniform on  $\{0, 1, 2\}$ . A randomized assignment independent of potential outcomes therefore gives the same distribution for (assignment, observed outcome); independent tasks give the same law at every sample size. In the first law,  $A$  wins for  $U = 0, 1$  and loses for  $U = 2$ , giving  $\theta_{\text{episode}} = 1/3$ . In the second,  $A$  wins for  $U = 0$  and loses for  $U = 1, 2$ , giving  $-1/3$ . For independent uniform draws the probability of either strict ordering is  $1/3$ , so both  $\theta_{\text{cross}}$  and the independent-replicate  $\theta_{\text{task}}$  given the constant observed context are zero under both laws.  $\square$

This is an identification obstruction, not a confidence-interval issue. Additional observations of the same randomized design cannot resolve  $\theta_{\text{episode}}$ . A binary single-tier net benefit is a special exception because it equals the difference of the two means; the proposition uses three ordered values. Running both systems in resettable task environments can identify the explicitly chosen clone/run coupling, subject to valid measurement and independent task sampling. Shadow runs do not reproduce live user reactions or downstream interference automatically.

#### A.4 A RANDOMIZED DISJOINT-PAIR DESIGN

Index nonoverlapping pairs of enrolled episodes by  $i$ . The pair has two positions,  $r = 1, 2$ , and potential outcomes  $Y_{ir}^A, Y_{ir}^B$ . Both positions are fixed before randomizing the orientation. All potential outcomes here refer to the fixed system versions, endpoint definitions, horizon, and resource caps. There is no interference between the two positions. Let  $R_i = 1$  assign  $A$  to position 1 and  $B$  to position 2;  $R_i = 0$  reverses these assignments. Let

$$P(R_i = 1 \mid \mathcal{H}_i, \{Y_{ir}^a\}_{r,a}) = q_i, \quad 0 < \varepsilon \leq q_i \leq 1 - \varepsilon,$$

where  $\mathcal{H}_i$  contains the information used to form and randomize the pair, and  $q_i$  is known and  $\mathcal{H}_i$ -measurable. This condition is satisfied by a fresh randomization coin with its recorded probability. It excludes assignments driven by unrecorded current outcomes.

For any measurable kernel  $\phi : \mathcal{Y}^2 \rightarrow [-1, 1]$ , define observed outcomes  $O_{i1}, O_{i2}$  and the score

$$Z_i^\phi = \frac{R_i}{2q_i} \phi(O_{i1}, O_{i2}) + \frac{1 - R_i}{2(1 - q_i)} \phi(O_{i2}, O_{i1}). \quad (13)$$

The arguments are always oriented as  $A$  then  $B$ . The within-pair symmetrized potential-outcome quantity is

$$m_i^\phi = \frac{1}{2} \{ \phi(Y_{i1}^A, Y_{i2}^B) + \phi(Y_{i2}^A, Y_{i1}^B) \}. \quad (14)$$

**Theorem 4** (Identification by pair randomization). *Under the stated randomization and consistency conditions,*

$$E(Z_i^\phi \mid \mathcal{H}_i, \{Y_{ir}^a\}_{r,a}) = m_i^\phi.$$

*For a filtration  $\mathcal{F}_{i-1}$  describing prior pairs and satisfying  $\mathcal{F}_{i-1} \subseteq \mathcal{H}_i$ , it follows that  $E(Z_i^\phi \mid \mathcal{F}_{i-1}) = E(m_i^\phi \mid \mathcal{F}_{i-1}) =: \mu_i^\phi$ . The score is bounded by  $|Z_i^\phi| \leq 1/(2\varepsilon)$ . With balanced orientation  $q_i = 1/2$ , it is simply the observed oriented kernel, and  $|Z_i^\phi| \leq 1$ .*

*Proof.* Under  $R_i = 1$ , consistency gives  $(O_{i1}, O_{i2}) = (Y_{i1}^A, Y_{i2}^B)$ . Its contribution to the conditional expectation is  $q_i \phi(Y_{i1}^A, Y_{i2}^B) / (2q_i)$ . Under  $R_i = 0$ , the contribution is  $(1 - q_i) \phi(Y_{i2}^A, Y_{i1}^B) / (2(1 - q_i))$ . Their sum is (14). The conditional-mean identity follows by iterated expectation. Exactly one term in (13) is nonzero, yielding the stated bound.  $\square$

Conditional exchangeability of the two positions is *not* needed for the symmetrized target in (14). If, additionally, the two potential-outcome vectors are conditionally independent draws from the same stratum-specific population, each term's conditional mean equals (12). Then balanced pairs estimate the independent-draw stratum preference. Pair formation by baseline stratum reduces cross-context comparisons but does not create same-task observations.

For a population component set  $\phi_j(y, y') = g_j(y) - g_j(y')$ . Algebra gives the particularly useful identity

$$m_i^{\phi_j} = \frac{1}{2} \sum_{r=1}^2 \{g_j(Y_{ir}^A) - g_j(Y_{ir}^B)\}. \quad (15)$$

Unlike a general hierarchy kernel, the component contrast is the average same-episode component effect for the pair. Thus the design supports population component gates and a cross-episode preference without incorrectly claiming a same-task preference effect.

If  $q_i \neq 1/2$ , omitting the inverse-probability factors generally targets  $q_i \phi(Y_{i1}^A, Y_{i2}^B) + (1 - q_i) \phi(Y_{i2}^A, Y_{i1}^B)$  instead. For example, if these two potential comparison signs are 1 and  $-1$  and  $q_i = 3/4$ , the symmetrized target is zero but the unweighted mean is  $1/2$ . The extension permits adaptive *orientation* probabilities; it still allocates one episode to each system per pair and is not a general bandit-router analysis.

#### A.5 ANYTIME CONFIDENCE SEQUENCES, INCLUDING DRIFT

The next result applies to any score sequence  $Z_i$ , including each hierarchy, tier, probability, or component score above. Let  $(\mathcal{F}_i)$  be a filtration to which the scores are adapted. Assume known  $\mathcal{F}_{i-1}$ -measurable bounds  $a_i \leq Z_i \leq b_i$ , set  $c_i = b_i - a_i$ , and write

$$\mu_i = E(Z_i \mid \mathcal{F}_{i-1}), \quad \bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mu_i, \quad \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n Z_i, \quad V_n = \frac{1}{4} \sum_{i=1}^n c_i^2.$$

The predictable bounds must hold before observing  $Z_i$ . If  $q_i$  depends on current contexts outside  $\mathcal{F}_{i-1}$ , the universal  $[-1/(2\varepsilon), 1/(2\varepsilon)]$  bound is valid; substituting a post-context narrower bound requires a filtration and centering argument that preserve predictability.

**Theorem 5** (Normal-mixture confidence sequence). *Fix  $\rho > 0$  and  $\alpha \in (0, 1)$  before examining this score stream. Define*

$$B_\alpha(v) = \sqrt{(v + \rho) \log\left(\frac{v + \rho}{\rho\alpha^2}\right)}, \quad C_n = \left[\hat{\mu}_n - \frac{B_\alpha(V_n)}{n}, \hat{\mu}_n + \frac{B_\alpha(V_n)}{n}\right]. \quad (16)$$

Then

$$P\{\bar{\mu}_n \in C_n \text{ for every } n \geq 1\} \geq 1 - \alpha.$$

One may intersect  $C_n$  with a known parameter range, such as  $[-1, 1]$ . For balanced pair scores in  $[-1, 1]$ ,  $V_n = n$ .

*Proof.* Let  $D_i = Z_i - \mu_i$  and  $S_n = \sum_{i \leq n} D_i$ . The conditional Hoeffding lemma gives, for every fixed  $\lambda \in \mathbb{R}$ ,

$$E\{e^{\lambda D_i} \mid \mathcal{F}_{i-1}\} \leq e^{\lambda^2 c_i^2 / 8}.$$

Therefore  $M_n(\lambda) = \exp\{\lambda S_n - \lambda^2 V_n / 2\}$  is a nonnegative supermartingale starting at one. Mix over  $\lambda \sim N(0, \rho^{-1})$ . Conditional Tonelli's theorem preserves the supermartingale property. Completing the square in the Gaussian integral gives

$$M_n = \sqrt{\frac{\rho}{V_n + \rho}} \exp\left\{\frac{S_n^2}{2(V_n + \rho)}\right\}.$$

Ville's inequality implies  $P(\exists n : M_n \geq 1/\alpha) \leq \alpha$ . If  $|S_n| \geq B_\alpha(V_n)$ , substitution gives  $M_n \geq 1/\alpha$ . Hence, outside an event of probability at most  $\alpha$ ,  $|\hat{\mu}_n - \bar{\mu}_n| < B_\alpha(V_n)/n$  for every  $n$ . This proves the claimed coverage. Intersecting with a set known to contain the target cannot remove it on the coverage event.  $\square$

No identical-distribution assumption appears in the theorem. Under drift, however, the target is the *running average conditional effect*  $\bar{\mu}_n$ , not the effect at the latest arrival or a future workload. If a predictable policy changes the stratum mix, hierarchy, or candidate version, that change is reflected in the sequence  $\mu_i$ . Predictability preserves the concentration argument but does not preserve a frozen deployment estimand. A new version or preference rule should ordinarily start a separately specified experiment.

The normal-mixture width scales as  $O(\sqrt{\log n/n})$  for bounded constant ranges and fixed  $\rho$ . It is convenient and explicit, but is not an iterated-logarithm-optimal boundary. Tuning  $\rho$  after examining this stream invalidates this particular guarantee unless that adaptation is accounted for.

#### A.6 BETTING TESTS FOR STATIONARY DEPLOYMENT CLAIMS

For stationary claims, a direct test can be more efficient than the range-based confidence sequence. Let  $Z_i \in [-1, 1]$  and test  $H_0 : E(Z_i | \mathcal{F}_{i-1}) \leq c$  for every  $i$ , where  $c \in (-1, 1)$  is fixed. Select  $J$  nonnegative stakes satisfying  $0 \leq \lambda_j < 1/(1+c)$  and mixture weights  $w_j > 0$ ,  $\sum_j w_j = 1$ , before examining the stream. Define

$$E_n(c) = \sum_{j=1}^J w_j \prod_{i=1}^n \{1 + \lambda_j(Z_i - c)\}. \quad (17)$$

These are standard bounded-mean betting tests (Waudby-Smith & Ramdas, 2024); sequential two-sample betting also has an established literature (Shekhar & Ramdas, 2024; Podkopaev & Ramdas, 2023).

**Theorem 6** (Validity and the finite-grid power condition). *Under  $H_0$ ,  $E_n(c)$  is a nonnegative supermartingale starting at one, and  $P\{\sup_n E_n(c) \geq 1/\alpha\} \leq \alpha$ . If the scores are i.i.d. under an alternative and at least one selected stake satisfies  $E \log\{1 + \lambda_j(Z_1 - c)\} > 0$ , then  $E_n(c) \rightarrow \infty$  almost surely. A fixed finite grid need not satisfy this condition for every alternative with  $EZ_1 > c$ .*

*Proof.* Each factor is positive because  $1 + \lambda_j(Z_i - c) \geq 1 - \lambda_j(1+c) > 0$ . Its conditional expectation is  $1 + \lambda_j\{E(Z_i | \mathcal{F}_{i-1}) - c\} \leq 1$ . Each product is therefore a nonnegative supermartingale; their fixed convex mixture is also one. Ville’s inequality proves the error bound. For a selected stake with positive expected log factor, the factors are bounded away from zero and infinity, so their logarithms are integrable. The strong law implies that the corresponding log product divided by  $n$  tends to its strictly positive expectation. That product and hence its positively weighted contribution to  $E_n(c)$  diverge. For the final claim take  $c = 0$  and a nondegenerate symmetric  $Z_1 \in \{-1, 1\}$ . Every fixed  $\lambda \in (0, 1)$  has negative mean log factor, since it equals  $\frac{1}{2} \log(1 - \lambda^2)$ . Continuity in the success probability implies that all finitely many positive grid stakes still have negative mean log factor for a sufficiently small positive mean. Zero stakes remain constant and cannot supply divergence.  $\square$

A countable positive-weight mixture with stakes tending to zero repairs the latter consistency limitation: for  $EZ_1 - c > 0$ , the derivative at zero of  $E \log\{1 + \lambda(Z_1 - c)\}$  is  $EZ_1 - c > 0$ , so all sufficiently small positive stakes have positive mean log factor. A reported finite grid retains exact error control, but its stake choices and empirical power must be reported without a universal consistency claim. The null in Theorem 6 is a conditional-mean null at every step. Merely requiring a nonpositive current cumulative mean under arbitrary drift does not imply the product’s supermartingale property. For that running-average target use Theorem 5.

#### A.7 CONJUNCTIVE DEPLOYMENT AND THE MULTIPLICITY DISTINCTION

Index the required criteria by  $j = 0, \dots, m-1$ . Criterion zero is hierarchical net benefit above a prespecified minimum  $c_0 \geq 0$ . Other criteria can be component noninferiority,  $\Delta_j > c_j = -\delta_j$ , with  $\delta_j \geq 0$ . A deployment claim is the conjunction that *every* criterion holds. Use the corresponding bounded scores from (13). Let  $L_{jn}(\alpha_j)$  be the lower endpoint in (16). The rule is

$$\tau = \inf\{n \geq 1 : L_{jn}(\alpha_j) > c_j \text{ for all } j\}. \quad (18)$$

The reported stationary betting implementation replaces each lower-bound condition with  $E_{jn}(c_j) \geq 1/\alpha_j$ : all criteria must pass at the same monitored look. An optional retained-evidence variant instead

uses  $\sup_{t \leq n} E_{jt}(c_j) \geq 1/\alpha_j$ . Retention is justified only for the fixed stationary claims; its stopping times and power differ from those of the reported current-look implementation.

**Theorem 7** (Stationary conjunction needs no alpha split). *Suppose each score has fixed conditional mean  $E(Z_{ij} | \mathcal{F}_{i-1}) = \mu_j$  and at least one requirement is false, namely  $\mu_{j_*} \leq c_{j_*}$ . Set  $\alpha_j = \alpha$  for every criterion. The confidence-sequence rule and both stationary betting variants have  $P(\tau < \infty) \leq \alpha$ , regardless of dependence among criteria.*

*Proof.* For the confidence-sequence rule, deployment requires  $L_{j_*n}(\alpha) > c_{j_*} \geq \mu_{j_*}$  at some  $n$ . Its probability is at most  $\alpha$  by Theorem 5. For betting, deployment requires the  $j_*$  process to cross  $1/\alpha$ ; its conditional-mean null holds, so Theorem 6 bounds this probability by  $\alpha$ . The index  $j_*$  can be chosen for each fixed data-generating law. No independence among criteria or union bound over them is needed.  $\square$

This is an intersection–union test for a *single conjunction*, not simultaneous confidence coverage for all component claims. It does not authorize choosing among multiple candidates, hierarchies, success routes, or deployment experiments at level  $\alpha$  each. For example, independent repeated experiments each allowed a level- $\alpha$  false deployment can have probability  $1 - (1 - \alpha)^M$  of at least one false deployment over  $M$  experiments. An experiment-wise allocation with  $\sum_e \alpha_e \leq \alpha_{\text{program}}$  gives a simple overall bound by the union inequality.

**Theorem 8** (Drifting running-average claims require simultaneous control). *Let  $\bar{\mu}_{jn} = n^{-1} \sum_{i \leq n} E(Z_{ij} | \mathcal{F}_{i-1})$ , with arbitrary allowed drift. If  $\sum_j \alpha_j \leq \alpha$ , the confidence-sequence rule satisfies*

$$P\{\exists n : \text{the rule deploys at } n \text{ and some } \bar{\mu}_{jn} \leq c_j\} \leq \alpha.$$

*Alternatively, level  $\alpha$  per criterion remains sufficient if there is one fixed index  $j_*$  such that  $\bar{\mu}_{j_*n} \leq c_{j_*}$  almost surely for every  $n$ .*

*Proof.* The union bound and Theorem 5 give probability at least  $1 - \sum_j \alpha_j$  that all lower bounds are valid for all indices and times. On this event deployment implies  $\bar{\mu}_{jn} \geq L_{jn} > c_j$  for every  $j$ , so the stated error event cannot occur. Under the fixed-index condition the error event is contained in failure of the one  $j_*$  lower confidence sequence, which has probability at most  $\alpha$ .  $\square$

Without a fixed violating component, a time-varying conjunction cannot use the stationary proof: the relevant failing component can change with  $n$ . Nor does a favorable historical average establish that a system will be safe after a workload shift. A finite-sample statement about the future needs a stability or transport assumption beyond the results here.

## A.8 DELAY, STRATA, AND VERSIONS

**Proposition 9** (Displaying completed enrollment prefixes). *Suppose the full episode scores in enrollment-pair order satisfy Theorem 5. At calendar time  $t$ , let  $N(t)$  be any possibly data-dependent integer such that the first  $N(t)$  pairs have their frozen complete endpoints available. Display  $C_{N(t)}$ , with no display at index zero. The same simultaneous coverage guarantee holds over all calendar display times.*

*Proof.* On the event that the confidence sequence covers at every integer  $n \geq 1$ , it covers at every selected integer  $N(t) \geq 1$ , simultaneously over  $t$ . Thus every calendar-time coverage failure is contained in the original failure event. No stopping-time property of  $N(t)$  is needed for this pathwise inclusion.  $\square$

The premise is substantive: the score’s conditional-mean and bound assumptions must hold in the filtration used for enrollment-order inference. Shared load, interference, and adaptive collection cannot be dismissed by invoking the delay proposition. A practical protocol fixes a horizon and treats failure to achieve a required outcome by that horizon as an endpoint, not as an observation to discard. It waits for each pair to mature and analyzes completed enrollment prefixes. Naively sorting by completion time or selecting only completed successes can change the comparison population. If endpoints remain genuinely missing at the fixed horizon, this analysis requires a prespecified

observed missingness outcome or a separate justified missing-data model; it does not impute validity to missing traces.

For a prespecified set of  $S$  strata with stationary parameters  $\mu_s$  and a fixed target mix  $w_s \geq 0$ ,  $\sum_s w_s = 1$ , construct within-stratum confidence sequences  $[L_{s,n_s}, U_{s,n_s}]$  at levels  $\alpha_s$  with  $\sum_s \alpha_s \leq \alpha$ . Set  $[L_{s,0}, U_{s,0}] = [-1, 1]$  before a stratum is sampled. Then

$$\left[ \sum_s w_s L_{s,n_s}, \sum_s w_s U_{s,n_s} \right] \quad (19)$$

covers  $\sum_s w_s \mu_s$  for every vector of sample counts with probability at least  $1 - \alpha$ . Indeed, simultaneous coverage of all stratum sequences implies the result by nonnegative weighting. Adaptive allocation of sample counts is allowed provided each within-stratum sequence retains its stated assumptions. Data-dependent weights selected to favor the candidate do not preserve a prespecified workload target. If both strata and gates require simultaneous drifting claims, allocate error over both indices.

#### A.9 OFFLINE REPEATED RUNS: THE TASK IS THE INDEPENDENT UNIT

Let  $X_i$ ,  $i = 1, \dots, n$ , be i.i.d. task clusters. The complete records consisting of task information, replicate counts, and all episode randomness are independent across  $i$ . Conditional on  $X_i$ , draw independent replicate sets  $Y_{ir}^A$ ,  $r = 1, \dots, R_{iA}$ , and  $Y_{is}^B$ ,  $s = 1, \dots, R_{iB}$ , with the stated system-specific laws. The positive replicate counts may be fixed or determined by task information before observing its outcomes. Define

$$T_i = \frac{1}{R_{iA}R_{iB}} \sum_{r=1}^{R_{iA}} \sum_{s=1}^{R_{iB}} h(Y_{ir}^A, Y_{is}^B), \quad \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n T_i. \quad (20)$$

**Theorem 10** (Task-level estimation and inference). *Under this design,  $E(T_i) = \theta_{\text{task}}$ ,  $T_i \in [-1, 1]$ , and  $\hat{\theta}_n$  is unbiased. If the entire task-and-replication records are also identically distributed across tasks, then*

$$\text{Var}(\hat{\theta}_n) = \frac{\text{Var}(T_1)}{n}.$$

*The normal-mixture confidence sequence with scores  $T_i$  and  $V_n = n$  is finite-sample valid in task order. Under the preceding i.i.d. complete-record condition, if  $\sigma^2 = \text{Var}(T_1) > 0$ , then*

$$\frac{\sqrt{n}(\hat{\theta}_n - \theta_{\text{task}})}{\{(n-1)^{-1} \sum_i (T_i - \hat{\theta}_n)^2\}^{1/2}} \implies N(0, 1).$$

*Proof.* Each replicate comparison has the conditional expectation in (11); averaging and iterating expectation proves unbiasedness. Boundedness follows because  $T_i$  is an average of numbers in  $[-1, 1]$ . Independence across tasks makes the variance of the mean equal to the sum of the task variances divided by  $n^2$ , which simplifies as stated under identical distribution. The bounded task scores meet Theorem 5. Finally, the ordinary i.i.d. central limit theorem applies because  $T_i$  is bounded. The sample variance converges to  $\sigma^2$  by the strong law, and Slutsky's theorem gives the studentized limit.  $\square$

Replicate pairs reuse episode outcomes and are dependent. Their number  $\sum_i R_{iA}R_{iB}$  is not an independent sample size. If tasks share a repository, user, or scenario with residual dependence, independent clusters at that higher level must replace individual tasks. If the benchmark is a fixed list rather than a sample of tasks, generalization to an external workload does not follow from a task bootstrap; one must state a fixed-list target or justify task sampling. Repeated runs do not repair that generalization gap.

For  $\text{WR} = p_W/p_L$ , use two task-level scores formed by replacing  $h$  with  $\mathbf{1}\{h = 1\}$  and  $\mathbf{1}\{h = -1\}$ . Simultaneous intervals  $[l_W, u_W]$  and  $[l_L, u_L]$  with  $l_L > 0$  imply  $\text{WR} \in [l_W/u_L, u_W/l_L]$ . If the lower loss bound is zero, there is no finite upper ratio bound from these intervals. In contrast, any confidence interval for  $\theta$  transforms monotonically to win odds on  $(-1, 1)$  via Proposition 1.

**Lemma 11** (Shared seeds: the diagonal changes the target). *For a fixed task  $X = x$ , suppose the paired seed records  $W_r = (Y_r^A, Y_r^B)$ ,  $r = 1, \dots, R$ , are i.i.d. conditional on  $x$ , but the two outcomes within  $W_r$  may be dependent. Let*

$$\theta_{\text{ind}}(x) = E\{h(Y_1^A, Y_2^B) \mid x\}, \quad \theta_{\text{coupled}}(x) = E\{h(Y_1^A, Y_1^B) \mid x\}.$$

For  $R \geq 2$ , the off-diagonal statistic

$$U_R(x) = \frac{1}{R(R-1)} \sum_{r \neq s} h(Y_r^A, Y_s^B)$$

is conditionally unbiased for  $\theta_{\text{ind}}(x)$ . The all-pairs statistic, including diagonal comparisons, satisfies

$$E\left\{\frac{1}{R^2} \sum_{r,s} h(Y_r^A, Y_s^B) \mid x\right\} = \left(1 - \frac{1}{R}\right) \theta_{\text{ind}}(x) + \frac{1}{R} \theta_{\text{coupled}}(x).$$

Its absolute bias for  $\theta_{\text{ind}}(x)$  is at most  $2/R$ .

*Proof.* For  $r \neq s$ , conditional independence of seed records gives  $E\{h(Y_r^A, Y_s^B) \mid x\} = \theta_{\text{ind}}(x)$ . For  $r = s$ , the expectation is  $\theta_{\text{coupled}}(x)$ . There are  $R(R-1)$  off-diagonal terms and  $R$  diagonal terms. Averaging proves both identities. Since both expectations lie in  $[-1, 1]$ , their difference divided by  $R$  has absolute value at most  $2/R$ .  $\square$

This is the usual finite-sample distinction between a U-statistic and its diagonal-including V-statistic, applied to paired stochastic agent runs. Averaging  $U_R(X_i)$  over independent tasks retains the bounded task-level inference in Theorem 10. The  $R(R-1)$  comparisons still are not independent observations. For  $R = 4$ , the all-pairs expectation contains one-quarter coupled-seed preference. With only one seed, no off-diagonal estimate is available. Removing diagonal pairs repairs within-seed coupling only under independence of the distinct seed records; shared infrastructure or other cross-seed dependence needs its own treatment.

#### A.10 SAMPLING COST OF REQUIRING EVERY GATE

**Proposition 12** (A finite-time tail bound for deployment delay). *Suppose stationary criterion  $j$  has conditional mean  $\mu_j = c_j + \Delta_j$  with  $\Delta_j > 0$  and constant score-range width  $d_j$ . Write  $r_{jn} = B_{\alpha_j}(nd_j^2/4)/n$  for its confidence radius. For any deterministic  $n$  satisfying  $r_{jn} < \Delta_j$  for all  $j$ , the confidence-sequence deployment time in (18) satisfies*

$$P(\tau > n) \leq \sum_{j=0}^{m-1} \exp\left\{-\frac{2n(\Delta_j - r_{jn})^2}{d_j^2}\right\}.$$

In particular, for fixed positive gaps and bounded widths, deployment occurs almost surely as data continue to accrue.

*Proof.* If every  $L_{jn} > c_j$ , the rule deploys by  $n$ . For a given criterion,  $L_{jn} \leq c_j$  implies  $\hat{\mu}_{jn} - \mu_j \leq -(\Delta_j - r_{jn})$ . The conditional Hoeffding lemma followed by the Chernoff bound gives probability at most  $\exp\{-2n(\Delta_j - r_{jn})^2/d_j^2\}$  for that event. The union bound proves the result without requiring independence between criteria. Because  $r_{jn} \rightarrow 0$ , the right side tends to zero. The events  $\{\tau > n\}$  decrease to  $\{\tau = \infty\}$ , so continuity of probability proves almost sure finite deployment.  $\square$

This bound is deliberately conservative. It makes explicit that a criterion close to its margin can dominate the sample requirement even when hierarchical preference is overwhelmingly favorable.

**Proposition 13** (A guardrail has an unavoidable information cost). *Consider a submodel in which each pair reveals one A failure indicator  $F_i \sim \text{Bernoulli}(p)$  independently, B never fails, and all other recorded information has the same conditional law under the hypotheses and gives no additional information about  $p$ . Fix a permitted failure rate  $\delta \in (0, 1)$  and an acceptable alternative  $p_1 \in (0, \delta)$ . Any sequential procedure with almost surely finite decision time  $T$  under both laws,  $E_{p_1}T < \infty$ , and deployment event  $D$  satisfying  $P_\delta(D) \leq \alpha < 1 - \beta \leq P_{p_1}(D)$  must obey*

$$E_{p_1}T \geq \frac{\text{kl}(1 - \beta, \alpha)}{\text{kl}(p_1, \delta)}, \quad \text{kl}(u, v) = u \log \frac{u}{v} + (1 - u) \log \frac{1 - u}{1 - v}.$$

*Proof.* Let  $\ell_i$  be the log likelihood ratio of one failure observation under  $p_1$  versus  $\delta$ . The stopped transcript likelihood ratio is  $\exp(\sum_{i=1}^T \ell_i)$ , since the procedure’s randomization mechanism is the same under both laws. Its relative entropy under  $p_1$  is

$$E_{p_1} \sum_{i=1}^T \ell_i = \sum_{i \geq 1} E_{p_1} [\mathbf{1}\{T \geq i\} \ell_i] = E_{p_1} T \text{kl}(p_1, \delta).$$

The interchange is valid because  $\ell_i$  is bounded for interior  $p_1, \delta$  and  $E_{p_1} T < \infty$ . The final equality uses the independence of  $F_i$  from the pre- $i$  stopping decision. Mapping the transcript to  $\mathbf{1}_D$  cannot increase relative entropy (the log-sum inequality), so this quantity is at least  $\text{kl}\{P_{p_1}(D), P_\delta(D)\}$ . For  $u > v$ , binary relative entropy increases with  $u$  and decreases with  $v$ , as follows by differentiating its displayed formula. The stated probability bounds therefore give a lower bound of  $\text{kl}(1 - \beta, \alpha)$ , proving the claim.  $\square$

For  $p_1 = \delta - \gamma$  with  $\gamma \rightarrow 0$  and fixed interior  $\delta$ , Taylor expansion yields  $\text{kl}(\delta - \gamma, \delta) = \gamma^2 / \{2\delta(1 - \delta)\} + o(\gamma^2)$ . Thus the required expected sample size can scale as the inverse squared distance to the guardrail. The result is a standard information-bound argument applied to this deployment subproblem, not a claim of a new general sequential lower bound.

#### A.11 SENSITIVITY OF A PREFERENCE CLAIM TO ITS COMPARATOR

**Proposition 14** (Comparator disagreement bound). *For any two prespecified measurable comparators  $h, g \in [-1, 1]$  on the same comparison population,*

$$|Eh - Eg| \leq E|h - g| \leq 2P(h \neq g).$$

*If  $Eh > 2P(h \neq g)$ , then  $Eg > 0$ . The factor two is sharp.*

*Proof.* The first inequality is the triangle inequality. The second follows pointwise because  $|h - g| \leq 2\mathbf{1}\{h \neq g\}$ . The sign claim follows from  $Eg \geq Eh - E|h - g|$ . Equality in both bounds is obtained when  $h = 1, g = -1$  on an event of probability  $q$  and  $h = g = 0$  otherwise, giving a difference  $2q$ .  $\square$

This supplies an interpretable sensitivity diagnostic when a tolerance, judge, or priority ordering changes: report the fraction of comparisons whose decision changes and whether they reverse or become tied. Statistical uncertainty in an estimated disagreement mass still needs inference. Choosing a favorable hierarchy after seeing the results is multiple selection, not prespecification. A finite family can be handled with simultaneous confidence sequences allocated over the family, or a separate validation stream; a result for one frozen hierarchy does not validate arbitrary post hoc searching.

#### A.12 SCOPE OF THE ESTABLISHED GUARANTEES

The proofs establish identification for the stated paired design, finite-sample error control for its bounded scores, and correctly defined deployment conjunctions. They do not establish that an unvalidated grader measures the intended construct; that an absence of observed rare harms certifies safety; that a benchmark represents a future workload; or that disjoint-pair analysis is statistically efficient relative to a correctly analyzed all-pairs U-statistic. Disjoint pairs intentionally simplify sequential dependence at a possible efficiency cost. Those issues require measurement audits, empirical comparisons, and application-specific assumptions rather than additional algebra on the same win score.

## B CONTINUOUS EVALUATION FROM PARTIALLY OBSERVED EPISODES

Waiting for the first unresolved episode can prevent a completed-prefix analysis from using many later outcomes. Selecting only completed episodes can instead introduce outcome-dependent selection. Here we retain the enrollment order but replace each unresolved score by guaranteed bounds. This is a worst-case completion construction: it does not require a model for how quickly favorable and unfavorable outcomes become observable. It specializes established full-data anytime inference and is closely related to the pending-outcome stopping construction of Henzi & Ziegel (2022b;a).

## B.1 LATENT ENROLLMENT ORDER AND THE ACTUAL OBSERVATION PROCESS

Let  $X_i$  be the complete bounded score of enrolled pair  $i$ , defined under the frozen evaluation horizon and protocol. It may be the observed oriented hierarchy score, its inverse-probability-weighted version from Theorem 4, or a component difference. The enrollment filtration  $(\mathcal{F}_i)$  used for the full-score argument need not be the calendar-time filtration actually observed by the evaluator. We require, rather than infer from the observation schedule, that

$$X_i \text{ is } \mathcal{F}_i\text{-measurable,} \quad a_i \leq X_i \leq b_i, \quad \mu_i = E(X_i \mid \mathcal{F}_{i-1}),$$

where  $a_i, b_i$  are known predictable bounds. Put  $V_n = \sum_{i \leq n} (b_i - a_i)^2 / 4$  and  $\bar{\mu}_n = n^{-1} \sum_{i \leq n} \mu_i$ .

At calendar time  $t$ , let  $N(t)$  be the finite number of fully enrolled and randomized pairs. The evaluator’s information  $\mathcal{G}_t$  contains whatever traces and labels have arrived by then. For each  $i \leq N(t)$ , it supplies measurable bounds  $\ell_i(t), u_i(t)$  satisfying

$$a_i \leq \ell_i(t) \leq X_i \leq u_i(t) \leq b_i \quad \text{simultaneously for every enrolled } i \text{ and every } t. \quad (21)$$

Bounds collapse to the score when it is determined. Outcome-dependent delay, out-of-order updates, and partial information are allowed. The observation process does not need to make  $X_i$  independent of its reveal time. For continuous calendar time, assume the relevant suprema and events are measurable; an event-driven implementation with countably many recorded updates suffices.

Condition (21) is a simultaneous guarantee about the *frozen complete score*. A point prediction, an ordinary pointwise 95% interval, or an unvalidated judgment of likely success does not satisfy it. If the guarantee instead holds on an event  $\mathcal{A}$  with  $P(\mathcal{A}) \geq 1 - \eta$ , all the error bounds below increase by at most  $\eta$ , without requiring independence.

**A sufficient experimental model.** One transparent model samples independent full potential episode records and their partial-reveal paths for each pair, uses fixed candidate systems and a fixed outcome protocol, and has no cross-pair interference. The record for pair  $i$  can include arbitrarily dependent final outcomes and reveal times. Pair orientation is then drawn with a fresh recorded randomization coin; its probability can depend on information from previously enrolled pairs. If that information is a function of their full records, it is measurable in the filtration revealing those records in enrollment order. The pair-score identification and bounded-score arguments apply in this filtration. Balanced orientation with a fixed independent workload supplies the stationary case. In a design-based version, full potential records and reveal paths can be conditioned upon, with independent assignment coins supplying the randomness.

These are sufficient conditions, not a claim that any asynchronous production system satisfies them. In particular, revealing all earlier final records in a mathematical filtration can change a conditional-mean restriction. Shared infrastructure effects, interference, or adaptive candidate updates require their own justification. The hierarchy must not be refitted after a pending episode’s partial outcome is seen. Time-uniform coverage and threshold-crossing guarantees transfer by pathwise event inclusion; free confidence-sequence and p-process lifting are established in Choe & Ramdas (2024). Their delayed-forecast discussion is also a close structural precedent. Such a transfer does not establish a calendar-time e-process or the required latent enrollment-order conditional-mean restriction.

## B.2 CONFIDENCE SEQUENCES FROM ALL ENROLLED PAIRS

Let  $B_\alpha(v)$  be the normal-mixture boundary in Theorem 5. For any  $1 \leq n \leq N(t)$  define

$$L_{n,t} = \frac{\sum_{i=1}^n \ell_i(t) - B_\alpha(V_n)}{n}, \quad U_{n,t} = \frac{\sum_{i=1}^n u_i(t) + B_\alpha(V_n)}{n}. \quad (22)$$

In particular,  $n = N(t)$  uses every enrolled pair, including unresolved pairs before or after a completed pair. The bounds use  $n$ , not the number of completed episodes, in both the sum and the boundary.

**Theorem 15** (Pathwise transfer of the full-score confidence sequence). *Under the full-score assumptions of Theorem 5 and the simultaneous enclosures (21),*

$$P\{L_{n,t} \leq \bar{\mu}_n \leq U_{n,t} \text{ for every } t \text{ and every } 1 \leq n \leq N(t)\} \geq 1 - \alpha.$$

*If the enclosures hold only on  $\mathcal{A}$ , the lower bound is  $1 - \alpha - \eta$ .*

*Proof.* Let  $\mathcal{C}$  be the event of full-score coverage for every integer  $n$ , which has probability at least  $1 - \alpha$  by Theorem 5. On  $\mathcal{C}$ , for every  $n$ ,

$$\frac{\sum_{i \leq n} X_i - B_\alpha(V_n)}{n} \leq \bar{\mu}_n \leq \frac{\sum_{i \leq n} X_i + B_\alpha(V_n)}{n}.$$

On the enclosure event, replacing the left sum by its lower bound and the right sum by its upper bound only widens this interval. This is a pathwise statement for every available  $(n, t)$ , so there is no need to assert that  $N(t)$  or a display time is an  $\mathcal{F}$ -stopping time. The desired event contains  $\mathcal{C}$  when the enclosures are certain, and contains  $\mathcal{C} \cap \mathcal{A}$  otherwise. A union bound gives the second probability statement.  $\square$

The width has an exact decomposition:

$$U_{n,t} - L_{n,t} = \frac{2B_\alpha(V_n)}{n} + \frac{1}{n} \sum_{i=1}^n \{u_i(t) - \ell_i(t)\}. \quad (23)$$

The second term is the price of unresolved outcome information. If  $X_i \in [-B, B]$  and only  $M$  of the first  $n$  scores remain unresolved, its contribution is at most  $2BM/n$  when the enclosures are clipped to  $[-B, B]$ . Thus one unresolved early pair need not prevent learning from arbitrarily many later pairs. This is a bound on information loss, not a guarantee that the method always produces a narrower interval than waiting for a smaller completed prefix; those procedures can also concern different running targets.

### B.3 PARTIAL-SCORE BETTING EVIDENCE

For every fixed- $B$  result below, clip the predictable bounds and all enclosures to  $[-B, B]$ , so that  $-B \leq a_i \leq \ell_i(t) \leq X_i \leq u_i(t) \leq b_i \leq B$ . Clipping preserves containment. For clarity use a fixed bound  $X_i \in [-B, B]$  and a fixed gate threshold  $c \in (-B, B)$ . Choose  $J$  stakes  $0 \leq \lambda_k < 1/(B+c)$  and weights  $w_k \geq 0$  with  $\sum_k w_k = 1$  before viewing the stream. The full-score mixture is

$$E_n^{\text{full}}(c) = \sum_{k=1}^J w_k \prod_{i=1}^n \{1 + \lambda_k(X_i - c)\}.$$

The observable partial-score lower wealth for any enrolled prefix is

$$\underline{E}_{n,t}(c) = \sum_{k=1}^J w_k \prod_{i=1}^n \{1 + \lambda_k(\ell_i(t) - c)\}. \quad (24)$$

Past factors are recomputed or updated when an enclosure improves; they are not counted as fresh independent observations. Stakes and mixture weights remain the ones committed before seeing this stream.

**Theorem 16** (Threshold validity of partial-score lower wealth). *If  $E(X_i \mid \mathcal{F}_{i-1}) \leq c$  for every  $i$ , then*

$$P\{\exists t, 1 \leq n \leq N(t) : \underline{E}_{n,t}(c) \geq 1/\alpha\} \leq \alpha.$$

*If the enclosures hold only on  $\mathcal{A}$ , the bound is  $\alpha + \eta$ . This establishes a valid anytime threshold test. It does not assert that  $\underline{E}_{N(t),t}(c)$  is a martingale or an  $e$ -process in  $(\mathcal{G}_t)$ .*

*Proof.* Under the null, each full-score factor is positive and has conditional expectation at most one. Hence  $E_n^{\text{full}}(c)$  is a nonnegative supermartingale starting at one, by the same argument as Theorem 6. On the enclosure event, nonnegative stakes give

$$0 < 1 + \lambda_k(\ell_i(t) - c) \leq 1 + \lambda_k(X_i - c).$$

Products and their nonnegative mixtures preserve the inequality, so  $\underline{E}_{n,t}(c) \leq E_n^{\text{full}}(c)$  for all  $(n, t)$ . Every partial-wealth threshold crossing therefore implies a crossing of the full-score process at some integer  $n$ . Ville’s inequality bounds the latter event by  $\alpha$ . A possible enclosure failure adds at most  $\eta$ . Nothing in this event inclusion proves a calendar-time conditional-expectation identity, which explains the final qualification.  $\square$

If desired, the process

$$p_t = \min \left\{ 1, \left[ \sup_{s \leq t} \max_{1 \leq n \leq N(s)} \underline{E}_{n,s}(c) \right]^{-1} \right\}$$

is an anytime-valid p-value process under exact enclosures, with an empty maximum interpreted as one. The threshold statement follows directly from Theorem 16. Calling its reciprocal an e-process would require an additional expectation guarantee and is not warranted by this argument alone.

**Corollary 17** (A prefix envelope contains the completed-prefix test). *Define  $\underline{E}_t^{\max}(c) = \max_{1 \leq n \leq N(t)} \underline{E}_{n,t}(c)$ . It has the same threshold guarantee as Theorem 16, without dividing  $\alpha$  over prefixes. If  $m(t)$  is a completely resolved enrollment prefix with  $\ell_i(t) = u_i(t) = X_i$  for  $i \leq m(t)$ , then*

$$\underline{E}_t^{\max}(c) \geq E_{m(t)}^{\text{full}}(c).$$

*Consequently the envelope test cannot cross later than the completed-prefix test with the same full process and monitoring times.*

*Proof.* The threshold event is exactly a subset of the event already bounded in Theorem 16. At the resolved prefix  $m(t)$ , partial and full factors coincide, and this prefix is one of the maximands. The crossing-time ordering follows pointwise.  $\square$

This corollary concerns the envelope, not merely the all-enrolled quantity  $\underline{E}_{N(t),t}$ . Adding many unresolved pairs can reduce that latter quantity, so the all-enrolled implementation has no universal speed ordering relative to completed-prefix inference. If each enclosure narrows over calendar time, a fixed-prefix lower wealth is nondecreasing as information arrives. Revisions that widen a previously overconfident enclosure indicate that its claimed pathwise guarantee was not valid.

For stationary means, the maximum over enrolled prefixes of  $L_{n,t}$  in (22) is also a valid lower bound on the same constant  $\mu$ . Under drift,  $\bar{\mu}_n$  depends on  $n$ , so maximizing lower bounds across prefixes does not produce a lower bound for  $\bar{\mu}_{N(t)}$ . The drifting analysis should retain the selected prefix and its corresponding target explicitly.

#### B.4 CONSTRUCTING GUARANTEED HIERARCHY ENCLOSURES FROM TRACES

Let  $\mathcal{C}_i^A(t)$  and  $\mathcal{C}_i^B(t)$  contain all final episode records consistent with the information observed for the assigned  $A$  and  $B$  episodes. Require that the actual complete records belong to these sets simultaneously. For the unweighted hierarchy define

$$\ell_i^h(t) = \inf_{a \in \mathcal{C}_i^A(t), b \in \mathcal{C}_i^B(t)} h(a, b), \quad u_i^h(t) = \sup_{a \in \mathcal{C}_i^A(t), b \in \mathcal{C}_i^B(t)} h(a, b). \quad (25)$$

These are bounds on the observed oriented comparison, not on an unobserved alternative-arm outcome for the same user. Measurability of the bounds is required; finite outcome-state enumeration is sufficient for binary tiers and possible signs of the resource comparison. The true final records are among the feasible completions, proving (21). A superset of feasible completions gives conservative bounds even if the separate completion sets ignore some within-pair constraints.

For a component score  $g(a) - g(b)$ , if the traces guarantee  $g(a) \in [g_A^-, g_A^+]$  and  $g(b) \in [g_B^-, g_B^+]$ , use

$$\ell_i^g = g_A^- - g_B^+, \quad u_i^g = g_A^+ - g_B^-.$$

For the inverse-probability pair score, multiply both endpoints by the known positive factor  $1/(2q_i)$  or  $1/\{2(1 - q_i)\}$  corresponding to the realized orientation. The resulting global or predictable score range must also be used in the full-score bound.

The hierarchy gives useful early conclusions without requiring every endpoint to be finalized. If all higher-priority tiers are guaranteed tied and the next tier is guaranteed decisive, its sign is final even when lower-priority resource records remain open. A stronger implication can sometimes combine several possible completions. For example, use success then cost with joint failures tied. Suppose  $A$  has completed with verified success at cost 1, while  $B$  is unresolved but has already incurred cost 2. Under a 5% relative cost tolerance and nondecreasing recorded cost,  $A$  wins if  $B$  ultimately fails and also wins if  $B$  ultimately succeeds: any final  $B$  cost is at least 2. Therefore  $\ell_i^h = u_i^h = 1$

before  $B$  completes. The success-difference enclosure is only  $[0, 1]$ , so superiority in success has not been established by that episode. A higher-priority safety tier still unresolved would invalidate this particular early win certificate.

Absence of a violation so far is not a certificate that the final episode is violation-free. Similarly, accumulated cost is a lower bound only when the recorded cost cannot later decrease. Refunds, post hoc verifier changes, or continuing activity after an apparently successful termination require broader completion sets. If a partial trace is missing, its possible completions remain in the bounds. Evaluation cannot improve evidence by deleting that pair.

#### B.5 WHEN CAN PARTIAL INFORMATION EVENTUALLY SUFFICE?

**Proposition 18** (Vanishing unresolved information). *Suppose  $X_i \in [-B, B]$  have fixed conditional mean  $\mu$ , and use a fixed normal-mixture tuning parameter. Along any observed sequence of calendar looks  $t_k$  with  $n_k = N(t_k) \rightarrow \infty$ , suppose*

$$\frac{1}{n_k} \sum_{i \leq n_k} \{u_i(t_k) - \ell_i(t_k)\} \longrightarrow 0 \quad \text{almost surely.}$$

*Then both bounds  $L_{n_k, t_k}$  and  $U_{n_k, t_k}$  converge to  $\mu$  almost surely. Thus a finite collection of stationary criteria with strict positive gaps above their thresholds eventually passes the partial-score confidence-bound rule, provided this condition holds for every criterion.*

*Proof.* Bounded martingale differences give  $n^{-1} \sum_{i \leq n} (X_i - \mu) \rightarrow 0$  almost surely. One direct proof uses the exponential fixed-time Hoeffding bound, its summability for any fixed positive deviation, and Borel–Cantelli, followed by a countable sequence of deviations tending to zero. By enclosure,

$$0 \leq \frac{1}{n_k} \sum_{i \leq n_k} \{X_i - \ell_i(t_k)\} \leq \frac{1}{n_k} \sum_{i \leq n_k} \{u_i(t_k) - \ell_i(t_k)\},$$

and likewise for  $u_i(t_k) - X_i$ . Hence the averages of lower and upper endpoints converge to  $\mu$ . The normal-mixture radius tends to zero because  $V_n \leq nB^2$ . The finite-gate conclusion follows by taking the maximum of the almost surely finite eventual-passing indices.  $\square$

The condition permits an unresolved first pair forever if the total unresolved width is  $o(n)$ . It can also hold when complete traces remain open but their hierarchy signs have already been certified. Conversely, if all scores remain enclosed only by  $[-B, B]$ , the lower bound cannot establish a positive mean without additional information. Robustness to informative delay is obtained through this explicit uncertainty cost.

**Proposition 19** (A bound on the loss of log wealth). *For a fixed positive stake  $\lambda < 1/(B + c)$  let  $\gamma = 1 - \lambda(B + c) > 0$ . For any  $n \leq N(t)$ , the corresponding full and partial single-stake log products satisfy*

$$0 \leq \log E_n^{\text{full}, \lambda} - \log \underline{E}_{n, t}^\lambda \leq \frac{\lambda}{\gamma} \sum_{i=1}^n \{X_i - \ell_i(t)\}.$$

*If the full-score sequence is i.i.d., the stake has strictly positive expected log growth, and the average upper-minus-lower width vanishes along  $n_k = N(t_k) \rightarrow \infty$ , its partial log product divided by  $n_k$  has the same positive limit. A fixed mixture assigning positive weight to that stake therefore eventually crosses every finite threshold.*

*Proof.* The function  $f(x) = \log\{1 + \lambda(x - c)\}$  is increasing on  $[-B, B]$  and has derivative at most  $\lambda/\gamma$ . Apply the mean value theorem to each  $f(X_i) - f(\ell_i(t))$  and sum. The strong law gives the full log-growth limit; the normalized difference vanishes by the width assumption and the displayed inequality. A fixed positive mixture weight contributes only a constant to the logarithm of its term.  $\square$

## B.6 DEPLOYMENT CONJUNCTIONS AND THE SCOPE OF THE GUARANTEE

Under fixed conditional means, every required stationary gate can use its partial-score threshold test at level  $\alpha$ . If one fixed gate null is true, an eventual deployment requires that gate’s partial wealth to cross, whose probability is at most  $\alpha$  by Theorem 16. Gates may cross at different times or prefix lengths because they concern fixed stationary parameters. This is the same intersection–union argument as Theorem 7, and does not require independence between reveal times or criteria. An enclosure failure event common to all gates adds at most  $\eta$ .

For arbitrary drifting running means, use the bounds (22) for the same stated enrolled prefix, allocate the confidence-sequence error across all required criteria, and require every lower bound to exceed its threshold. The union-bound proof of Theorem 8 then applies pathwise. A historical partial-bound crossing cannot be retained as evidence about a different current drifting target. The same candidate-selection and repeated-test restrictions continue to apply.

These results address eventual outcomes through a *fixed episode horizon*. They are distinct from the calendar-time cumulative reward target studied by Lindon & Kallus (2026), who analyze delayed randomized outcomes and their filtration difficulties. No assertion of calendar-time martingale structure is needed here; nor do our bounds contradict that work’s obstruction for a different, calendar-time treatment-effect process. Stopping enrollment does not authorize changing the endpoint definition for pending episodes. If deployment terminates pending runs, either the prespecified intervention includes that termination or the estimand remains the stated hypothetical continuation through the frozen horizon.

## C EXPERIMENTAL DETAILS AND FULL NUMERICAL OUTPUTS

The executable sources and CSV files accompanying this paper are the canonical numerical record. The primary synthetic data-generating parameters are listed below. Each pair independently draws two agents’ compliance and success indicators; costs have log standard deviation 0.45, with the indicated A/B scale ratio. The cost tie threshold is 5% of the larger cost. Success and compliance are distinct measurements: an episode can complete the task while violating policy.

| Scenario              | Compliance A | Compliance B | Success A | Success B | Cost A/B |
|-----------------------|--------------|--------------|-----------|-----------|----------|
| Identical             | .995         | .995         | .75       | .75       | 1.00     |
| Efficiency gain       | .995         | .995         | .75       | .75       | .55      |
| Success regression    | .995         | .995         | .65       | .75       | .40      |
| Compliance regression | .975         | .995         | .75       | .75       | .45      |
| Joint gain            | .995         | .995         | .84       | .75       | .75      |
| Weak gain             | .995         | .995         | .76       | .75       | 1.00     |

The initial simulation uses seed 20260917 with separate scenario streams. Each gate uses an equal mixture of 40 bets geometrically spaced from  $10^{-4}$  to  $0.99/(1+c)$ , where  $c$  is its threshold; no bet is selected by looking at the results. Normal-mixture bounds fix  $\rho = 100$ . Unadjusted and fixed-horizon Wald comparisons use one-sided 0.05 cutoffs. The ten-look group comparator uses a one-sided 0.005 cutoff at each look. It is a simple multiplicity-adjusted asymptotic baseline, not an implementation of a particular published group-sequential win-ratio covariance estimator. A capped stopping time equals 10,000 when no deployment occurs; conditional stopping times among deployments are reported separately in the CSV and must not be conflated with expected sample use.

The boundary, dependence, and adaptive-order extensions use seed 20260918. Boundary compliance and success are set to 0.985 versus 0.995 and 0.72 versus 0.75, respectively, with cost ratio 0.4. The dependence experiment generates a normal task-specific A–B contrast with standard deviation 1.2 and four independent unit-normal run errors per arm; its unconditional signed preference is zero by symmetry. Task-level standard errors use 39 degrees of freedom; the naive comparator divides the standard deviation of all pair scores by the square root of 640.

The adaptive-order experiment sets potential score means to 0.8 and  $-0.8$  for the two assignment orientations. At pair  $i$ , the order probability is  $0.1 + 0.8 \expit(1 + S_{i-1}/\sqrt{i})$ , where  $S_{i-1}$  sums past unweighted scores. This is predictable and obeys positivity. The inverse-probability score has

symmetric target zero and lies in  $[-5, 5]$ ; betting and normal-mixture bounds use the appropriate range. It has 1,000 repetitions, at most 3,000 pairs, and looks every ten pairs after pair 100.

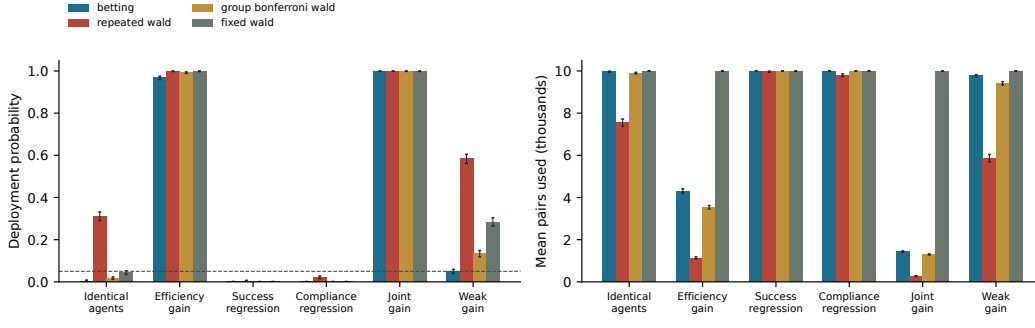


Figure 3: Observed deployment rates and capped mean sample use. Bars are Monte Carlo summaries, not evidence of universal ordering. Error bars are 95% Wilson intervals for rates and 1.96 Monte Carlo standard errors for mean pair counts. The nominal horizontal line applies to false deployment under the composite deployment null.

## D PUBLIC-DATA PROVENANCE AND INTERPRETATION

The  $\tau^2$  source repository is pinned to commit `b7ea9074c1cba482b30687fecdb5c8425fd6f619`. The archived runs predate that retrieval commit. The compared three models share run commit `c30d59a` in airline and retail and `f125e68` in telecom; source manifests retain full identifiers. The model snapshots are GPT-4.1-2025-04-14, o4-mini-2025-04-16, and Claude-3-7-sonnet-20250219. The fourth downloaded model, GPT-4.1-mini, used a different harness commit and was excluded from the primary comparison set before numerical analysis.

Task success uses the source’s binary `reward.info.reward`; historical agent cost uses `agent.cost`, independently reconciled to assistant-message costs. Tool-call count sums assistant tool calls. All primary source records have finite nonnegative cost and complete success values. No failed run is dropped to improve a system’s outcome. The  $\tau^2$  repository has since corrected tasks and evaluations; these archived labels are not certified against every later correction.

SWE-agent results use the two 2024-04-02 GPT-4 and Claude-3-Opus submissions on SWE-bench Lite. Resolved-task identifiers come from the official results files; per-instance cost uses `info.model_stats.instance_cost`, not cumulative total cost. Many trajectories terminate at a cost cap. This is part of the archived system configuration. The repository retrieval commit, S3 source URLs, and SHA256 digests of all 600 trajectories and result files are saved. Only numerical derivatives and retrieval manifests are included in this package; original patches and transcripts are not redistributed.

Bootstrap resampling preserves all runs of a task and the two systems together. Distinct benchmark tasks are treated as exchangeable units for this descriptive calculation. The analysis does not establish that benchmark tasks are an iid sample from production, nor that the four globally reused seeds generate independent task noise. The SWE leave-one-repository-out analysis is a stability diagnostic, not a replacement for a repository-cluster confidence interval. Consequently, inferences beyond the fixed historical workload require additional sampling and dependence assumptions.

The maintained CSV files include every primary contrast, cost tolerances 0, 0.10, and 0.20, reversed resource priority, common-seed pairing, and all-pairs sensitivity. The internally dated analysis plan and amendment are included. They are not a public preregistration. Some additional exploratory suggestions in the feasibility audit remain unexecuted; no conclusion relies on those suggested analyses.

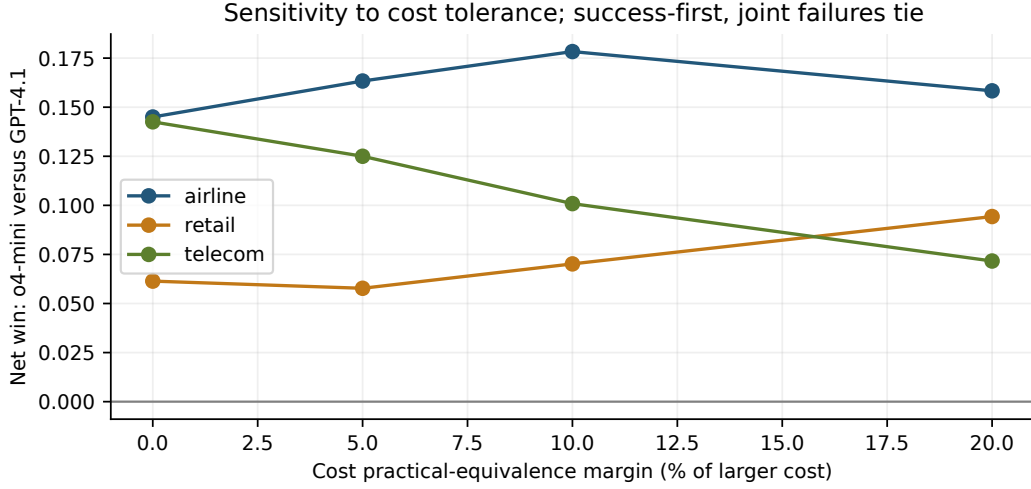


Figure 4: Cost-tolerance sensitivity under a fixed hierarchy. Changing the threshold changes the preference estimand. All settings are reported; none is selected because it favors a system.

## E DELAYED-FEEDBACK EXPERIMENT DETAILS

The master seed is 2026091803, with independent scenario child seeds. Success bits are independent Bernoulli(0.90), and positive costs are independent lognormal variables with log standard deviation 0.45, independent of success. The  $A/B$  scale ratio is  $r = 1$  or 0.55. The analytic success difference is zero and net benefit is

$$0.9^2 \left[ \Phi \left( \frac{\log(0.95) - \log r}{\sqrt{2}(0.45)} \right) - 1 + \Phi \left( \frac{-\log(0.95) - \log r}{\sqrt{2}(0.45)} \right) \right].$$

The null value is exactly zero; the alternative is approximately 0.52724. The hierarchy compares success first, then cost with 5% relative maximum-cost tolerance when both succeed; joint failures tie.

Pairs enroll at ticks 1–60 in batches of 100.  $A$ ’s success grade arrives one tick after enrollment if successful and 20 ticks later otherwise;  $B$ ’s grade always arrives after one tick.  $A$ ’s cost arrives after two ticks if at most 1 and after 20 ticks otherwise;  $B$ ’s cost arrives after two ticks. This outcome-dependent telemetry is identical across scenarios. The construction deliberately favors early favorable  $A$  measurements. No record remains permanently missing. The last look is tick 80.

A complete pair has both success grades and, only when both succeed, both costs. Thus irrelevant pending costs do not hold up the completed baseline. At each tick the valid methods consider the same deterministic prefix grid 100, 150,  $\dots$ , 6000. Both gates must cross 20 at one common prefix. Complete-prefix inference considers every grid prefix with all required scores determined, not just the latest prefix. The partial method additionally considers enrolled prefixes with unresolved scores. Each gate uses 40 equal-weight geometric bets from  $10^{-4}$  to  $0.99/(1+c)$ , where  $c = 0$  or  $-0.03$ . All choices were recorded before execution; this was an internal dated protocol, not public preregistration.

For each missing success bit, enumerate both possible values. Component lower bounds are the smallest feasible success difference. The hierarchy lower bound is the minimum over these possible success pairs: discordant success determines the sign, joint failure ties, and joint success permits all three cost signs unless both costs are observed. These deliberately broad sets remain valid without using a model for the delay distribution. Observed timing is not treated as a perfect classifier of a hidden bit. The score bounds are ternary and are checked by exhaustive finite-state cases and samplewise containment against simulated complete outcomes.

Nondeployments receive the cap of 80 ticks and 12,000 initiated executions. Execution counts are twice the number enrolled before the hypothetical procedure stops; generated latent values after that stopping point are used solely for paired method comparison. The reported observed

cost sums measure telemetry completeness, not dollars paid or expenditure saved. The source, configuration, protocol and result hashes, numerical tables, paired gains, and complete deployment curves are supplied. The verification checks include partial-wealth domination, same-prefix gating, non-later stopping, and equal final decisions after full reveal. The finite simulation does not test the ongoing-enrollment consistency propositions, which require  $n \rightarrow \infty$  and vanishing unresolved width.

Table 2: Frozen delay study, 1,000 repetitions per row. Means cap nondeployments at 80 ticks and 12,000 initiated executions.

| Scenario       | Procedure                | Deployments | Mean ticks | Executions |
|----------------|--------------------------|-------------|------------|------------|
| Equal outcomes | Completed only (invalid) | 1000        | 4.155      | 831.0      |
| Equal outcomes | Complete prefixes        | 6           | 79.763     | 11976.6    |
| Equal outcomes | Partial prefixes         | 6           | 79.763     | 11976.6    |
| Cheaper $A$    | Completed only (invalid) | 1000        | 3.958      | 791.6      |
| Cheaper $A$    | Complete prefixes        | 997         | 41.285     | 8130.8     |
| Cheaper $A$    | Partial prefixes         | 997         | 22.363     | 4459.6     |

## F FEASIBLE CERTIFICATES FROM ARCHIVED AGENT TRAJECTORIES

**Design and information boundary.** This descriptive audit uses the same nine hashed  $\tau^2$  archives as the historical comparison: three models, 278 tasks, four trials, and all three fixed model contrasts. Twelve off-diagonal seed-index comparisons per task and contrast give 10,008 dependent comparisons. The analysis specification was frozen before the replay aggregates were computed, after the archive and its complete-outcome analysis were known. It is an internal protocol, not external preregistration or a newly sampled benchmark. We retain every domain/contrast combination.

At integer tick  $k$ , expose the  $k$ th actual assistant message of each arm, including its accrued cost and issued tool calls. Zero-cost initial messages remain in the sequence. The final success label, cost, and tool-call total become available at a terminal marker one tick after that arm’s last assistant message. This accommodates trailing records and final verification by convention; it is not an observed grading delay. The rule receives only revealed values. It never uses a pending episode’s final label, length, cost, or future messages to tighten bounds.

**Cost reconciliation and completion sets.** All 3,336 episodes and 51,247 assistant messages had finite nonnegative costs; 3,336 zero-cost initial messages were retained. Summed message costs agreed with final recorded agent cost within the specified tolerance; the largest absolute discrepancy was  $3.565 \times 10^{-16}$ . Decimal arithmetic uses the archived numeric strings. From every pending cumulative cost we subtract the same conservative allowance,  $7.446604 \times 10^{-10}$ , clipping at zero. This archive-wide maximum reconciliation allowance is fixed by the initial audit; it widens the sets and does not disclose a pending episode’s own final value. Cost and tool counts have no inferred upper bound. Pending success remains in  $\{0, 1\}$ .

Enumerating feasible success bits and resource signs gives a conservative interval for success first, then cost with 5% relative tolerance among joint successes, then issued tool calls. Two failures tie. There is no independent safety label or safety tier. If  $A$  has completed successfully and  $B$ ’s certified cost lower bound exceeds  $c_A/0.95$ , every completion favors  $A$ :  $B$  either fails, or succeeds at a losing cost. Such a certificate determines the sign without determining the eventual deciding tier. All 3,426 early certificates retained this ambiguity: 3,163 allowed success or cost, and 263 also allowed the tools tier. We do not report these as early verified successes or as known cost-tier wins. The supplied diagnostic examples contain only revealed values; retrospective pair records separately contain final tiers and lead times.

**Validation and interpretation.** All 195,171 prefix intervals contained the archived final sign, were nested, and collapsed after completion. Final Decimal signs matched the existing floating-point comparator on all 10,008 comparisons. A separate implementation reconstructed every prefix from the raw archives with a closed-form completion rule and reproduced all early resolution times. Independent perturbation checks confirmed that hidden final values and future suffixes do not alter a pending revealed state. Among the 3,426 early resolutions, 831 led only the terminal marker by one

Table 3: All archived trace-certificate comparisons. Percentages use the full row denominator. “Before last marker” means before the later terminal marker; “Before messages” additionally requires at least one actual assistant message to remain unseen. The difference measures sensitivity to the added terminal-marker convention. These are descriptive, dependent comparisons, not binomial trials.

| Domain  | Models $A/B$         | Pairs  | Before last marker (%) | Before messages (%) |
|---------|----------------------|--------|------------------------|---------------------|
| Airline | o4-mini / GPT-4.1    | 600    | 39.2                   | 27.2                |
| Airline | Claude-3.7 / GPT-4.1 | 600    | 36.3                   | 30.0                |
| Airline | o4-mini / Claude-3.7 | 600    | 41.7                   | 34.5                |
| Retail  | o4-mini / GPT-4.1    | 1,368  | 40.1                   | 26.4                |
| Retail  | Claude-3.7 / GPT-4.1 | 1,368  | 54.0                   | 41.1                |
| Retail  | o4-mini / Claude-3.7 | 1,368  | 32.7                   | 21.3                |
| Telecom | o4-mini / GPT-4.1    | 1,368  | 31.5                   | 26.8                |
| Telecom | Claude-3.7 / GPT-4.1 | 1,368  | 21.0                   | 16.7                |
| Telecom | o4-mini / Claude-3.7 | 1,368  | 19.7                   | 17.3                |
| All     | All three contrasts  | 10,008 | 34.2                   | 25.9                |

tick; 2,595 occurred while actual messages remained. This secondary descriptive split was added during independent review, without changing the frozen replay or selecting contrasts.

The completion-only comparator resolves at the later terminal marker, when both final labels and resources are known. Table 3 shows that useful completion bounds occur in real archived records, including the motivating retail and telecom contrasts. However, the ordinal schedule is artificial, original grader labels may be imperfect, and cross-run comparisons share tasks and trajectories. We perform no population confidence interval, sequential hypothesis test, deployment decision, or wall-clock saving calculation from these counts. A live evaluation must validate its own terminal semantics, irrevocable costs, grader errors, enrollment assumptions, and information available at each decision. These observations establish archival certificate feasibility, not operational latency or deployment benefit.

## G DECISION OBJECTIVES AND SUCCESS-GRADER SENSITIVITY

The protocol was frozen before execution of this additional study, after the original simulation study had been inspected. All ten cases use 500 independent repetitions, 6,000 pairs, master seed 2026091804, and looks at 100 and every 100 pairs thereafter. The same 40 positive bets and same-look conjunction are used. Fixed weighted utility is  $0.8S + 0.1C + 0.1/(1 + \text{cost})$ ; bounded efficiency is  $1/(1 + \text{cost})$ . Both are in  $[0, 1]$ , so their pair differences use the same bounded-score betting construction. Their cost scale and weights are illustrative preferences, fixed before seeing this study. Giving utility to cost on unsuccessful episodes differs from the hierarchical resource-eligibility rule.

The six compared decisions require respectively: success superiority; only success/compliance noninferiority; guarded hierarchical preference; utility superiority; guarded utility superiority; or guarded efficiency. An approval is not labeled a statistical error solely because another method’s different deployment region fails. Population Pareto labels use true success, compliance and raw mean cost; they are descriptive facts of the generating laws, not estimated confidence regions. Mean lognormal cost is analytic, and bounded-efficiency means use Gauss–Hermite quadrature with 80/160-node agreement checked.

**What can a known label-error guarantee protect?** For an iid episode-pair law let  $\tilde{S}$  be a measured success bit and suppose  $P(\tilde{S}^a \neq S^a) \leq \epsilon_a$  for  $a = A, B$ . All other metrics and the frozen comparison rule agree. Put  $\epsilon = \epsilon_A + \epsilon_B$ . A union bound and bounded scores give

$$|\tilde{\Delta}_S - \Delta_S| \leq \epsilon, \quad |\tilde{\theta} - \theta| \leq 2\epsilon.$$

Indeed a success difference can change by at most the sum of its two label-error indicators, and the hierarchical sign changes by at most two on the event that at least one label is wrong. Taking expectations proves both bounds. Hence measured superiority thresholds  $2\epsilon$  for net benefit and  $-0.03 + \epsilon$  for success, with unchanged compliance threshold, suffice for the true deployment region.

For the iid study, the corresponding fixed conditional nulls and intersection–union proof give the same anytime false-deployment bound. Under temporal adaptation, analogous conditional error bounds would need independent justification. Uncertainty in estimated grader error rates is not covered by treating those estimates as known constants.

The four noise cases use conditional false-positive/false-negative rates (FP,FN): identical agents with (.05,.05) in both arms; a true success harm (.71 versus .75) with  $A = (.05, 0)$  and  $B = (0, .05)$ ; an efficiency gain (.75 versus .75, cost ratio .55) with the adverse reverse pattern; and larger true harm (.65 versus .75) with (.05,.05) in both arms. For the harm cases the cost ratio is .4. All compliance probabilities are .995. Noise is independent of cost and compliance conditional on true success and arm. The measured success probability is  $p(1 - \text{FN}) + (1 - p)\text{FP}$ , which gives the analytic measured hierarchical target after substitution in the original target formula. The conservative armwise error bound is  $\max(\text{FP}, \text{FN})$ ; it may exceed the actual unconditional error frequency.

For favorable differential grading of the smaller harm, the 97.8% measured approval rate has Wilson 95% Monte Carlo interval [96.10%, 98.77%]. True-label and conservative-bound approvals are zero in 500 repetitions. With adverse differential labels on a real efficiency gain, true-label approval is 73.8% and measured approval is zero. These results concern the four stated noise laws. They do not validate a real judge, recover latent truth without assumptions, or establish power of the conservative correction. In all four cases, the assumed summed error bound is 0.10, making its measured-success threshold 0.07; every measured contrast is below that threshold. Thus all four are null cases for this sufficient rule, and no bound-aware alternative was studied. Zero approvals are not evidence of corrective power.

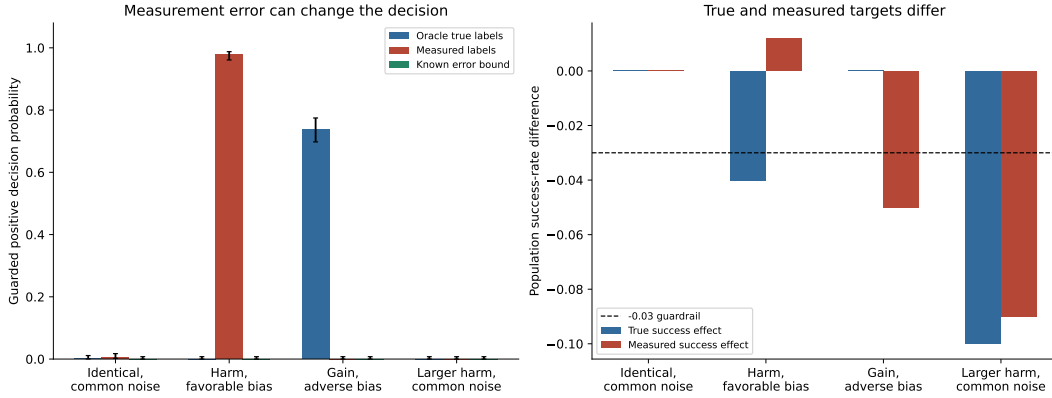


Figure 5: Success-grader sensitivity under four fixed noise laws. Oracle labels exist only in simulation; the bound-aware rule assumes externally valid error limits. Rates are Monte Carlo decisions about the indicated measured or true target, not evidence about an actual deployed grader.

## H PROSPECTIVE WORKFLOW PILOT

The pilot used the pinned  $\tau^2$ -bench source revision b7ea9074c1cb (full hash in the protocol) and its base telecom tasks. From tasks with deterministic action/environment criteria and no natural-language grading assertion, we selected the twelve lowest hashes of the master seed 20260918 and task identifier. The complete tasks, selection, randomized within-task workflow order, independent arm seeds, model identifiers and exact verification prompt were frozen in a dated local protocol before model outputs were collected. This is a laboratory workflow experiment with simulated users, not production-user A/B testing.

An initial protocol compared GPT-5.6-Luna and Claude Haiku 4.5, with a common Luna user simulator. Its first request failed for exhausted OpenAI quota, yielding no completed trajectory. The original protocol, zero-completion records and \$0.0078588 retained reservation remain immutable. Before collecting any completed output, we recorded an infrastructure-motivated amendment: standard versus verification-prompt workflows, both using claude-haiku-4-5-20251001, with the same Haiku user simulator in both arms. The twelve tasks and mapped randomized order were retained. No task, prompt, step or monetary limit was changed after observing the amended pilot’s outcomes.

The standard agent uses the upstream default system prompt and telecom policy. The additional verification instruction asks the agent to check the applicable policy and observed state before a state-changing tool call, verify corrective actions with available diagnostics or specific user confirmation, and avoid declaring resolution without an observable check. It explicitly does not override domain policy. Both workflows use temperature zero, at most 40 steps, four errors, output limits of 384 agent tokens and 192 user-simulator tokens, and one trial per task and workflow. Provider randomness is not guaranteed reproducible by the simulator seeds. API calls use direct transport with zero retries and no model fallback. A conservative reservation precedes each request; actual reported usage is then priced at the frozen list rates. Uncertain failed requests retain their reservation, and an unaffordable next request stops the experiment. These constraints are part of the evaluated workflow.

There were 484 completed amended-pilot requests. Eighteen trajectories completed on nine task pairs; the next trajectory stopped before a request could fit the remaining reservation budget, and the remaining planned records were retained as unobserved. Standard and verification each had two successes, seven completed failures, six step-cap terminations and three unobserved planned runs. Among the nine complete pairs, verification had zero wins, two losses and seven ties. Complete-case net preference  $-2/9$  is descriptive only: stopping on cumulative cost can select which tasks are completed. The full planned cohort therefore remains explicit.

If the known score sum over  $m$  of  $N = 12$  pairs is  $s$ , each unknown pair lies in  $[-1, 1]$ , giving deterministic support bounds  $[(s - (N - m))/N, (s + (N - m))/N]$ . With  $s = -2, m = 9$  these are  $[-5/12, 1/12]$  for the planned-cohort net preference. The observed success difference sum is zero, yielding  $[-3/12, 3/12]$ . These conservative bounds ignore partial one-arm information. They express missing finite-cohort outcomes, not population sampling uncertainty or a confidence sequence.

The amended pilot’s usage-priced list cost was \$3.937767. Including the retained initial reservation gives \$3.9456258 under the combined \$4 cap. Two separate provider diagnostics account for \$0.002035, so the total project ledger is \$3.9476608 under its initial \$5 allocation. These are conservative usage/reservation accounts, not reconciled provider invoices. Credentials and private raw text are excluded from release. Sanitized trace metadata, tool names, text hashes, token use, task checks, complete planned-run records, frozen source hashes, and the cost ledger permit inspection of the reported summaries.

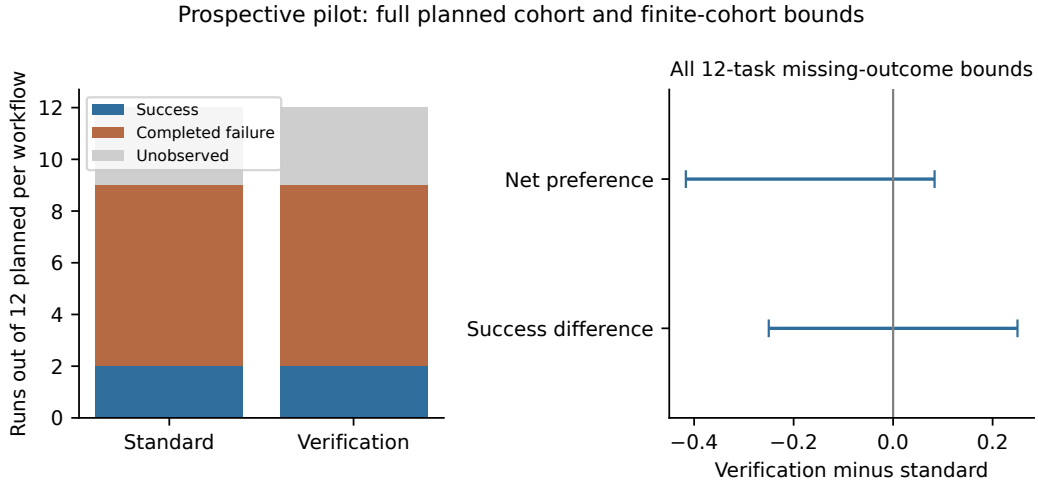


Figure 6: Prospective laboratory feasibility pilot. All planned tasks, including unobserved runs, remain in the cohort. Horizontal intervals are deterministic missing-outcome bounds, not sampling confidence intervals.

## I COMPLETE-DATA DIRICHLET-MIXTURE REFERENCE

For each raw gate score, let  $x_n = (w_n, t_n, l_n)$  be the counts of  $+1, 0, -1$  among iid pairs with fixed probability vector  $p$ . For fixed positive prior  $a = (1, 1, 1)$ , the known Dirichlet-mixture likelihood

ratio is

$$M_n(p) = \frac{B(a + x_n)}{B(a) \prod_j p_j^{x_{n,j}}}, \quad E_n^{\text{DM}}(c) = \inf_{p: p_w - p_l \leq c} M_n(p).$$

Here  $B$  is the multivariate beta function. These are established categorical-inference ingredients (Lindon & Malek, 2022). For an interior true vector  $p_0$ , each alternative categorical likelihood ratio is a mean-one nonnegative martingale. Integrating over the fixed Dirichlet prior preserves that property. At a boundary true vector, the likelihood ratio restricted to the true support is a nonnegative supermartingale, which suffices. For any null  $p_0$ , the composite statistic is pointwise at most  $M_n(p_0)$  at every  $n$ . Thus an eventual crossing of  $1/\alpha$  is included in the crossing event of this true-null supermartingale, giving probability at most  $\alpha$  by Ville’s inequality. The infimum need not itself be a martingale. Applying the fixed-null intersection–union argument to separate preference, success and compliance scores supplies the stationary guarded-decision guarantee without assuming independence between scores within a pair.

The logarithm uses the maximum null log likelihood. If the empirical categorical MLE satisfies  $p_w - p_l \leq c$ , use that MLE; otherwise concavity places the optimum on the boundary. Write  $p_l = u$ ,  $p_w = u + c$ ,  $p_t = 1 - 2u - c$ , where  $\max(0, -c) \leq u \leq (1 - c)/2$ . The interior score equation is

$$-2nu^2 + \{w(1 - c) + l(1 - 3c) - 2tc\}u + lc(1 - c) = 0.$$

Evaluate all feasible roots and endpoints, with zero-count terms  $0 \log 0 = 0$ , and select the maximum likelihood. The isolated implementation checks nonnegative integer counts, positive prior and valid thresholds. Independent constrained-optimization and small-sample exact-crossing checks accompany the code audit. General-purpose functional projection and confidence-width routines are not used for these results.

This likelihood argument assumes a fixed categorical vector; fixed conditional means alone are not the same assumption. It is used here only for complete iid scores, not adaptive importance-weighted counts or drifting targets. Decisive hierarchy cells do not recover raw success or compliance differences, so each required raw score has its own counts. The resulting intersection–union guarantee is for one deployment claim, not simultaneous confidence coverage for all effects.

The statistic also cannot replace the positive-factor lower-bound rule: at  $c = 0$ , counts (10, 0, 0) have log evidence 2.741817, whereas reducing one win to a tie yields (9, 1, 0) and log evidence 2.996915, exceeding log 20. Consequently lower scores need not yield lower DM evidence. An asynchronous DM extension would require its own feasible-completion infimum or a completed-prefix protocol.

The reference study uses seed 20260918, batches of 25, 2,000 repetitions per scenario and 10,000 pairs. It includes the original six laws and two post-primary-study extensions with success 0.30 in both arms: equal cost and cost ratio 0.55. All scenarios and methods are retained. This is an exact isolated reproduction of an existing contributed comparison, not a newly preregistered study. All 64 method/scenario rows reproduce exactly; paired sample-use differences and source hashes are supplied. Each method uses current-look conjunction, matching the original protocol. The different seed explains small differences from the primary simulation.

Table 4: Separate complete-data reference study: deployment percentages, 2,000 repetitions per row, same draws for all methods. The initial six laws retain their original definitions; the last two are extensions.

| Scenario              | Guarded bet | Guarded DM | Win-only DM | Repeated Wald |
|-----------------------|-------------|------------|-------------|---------------|
| Identical agents      | 0.60        | 0.00       | 0.00        | 29.90         |
| Efficiency gain       | 96.60       | 60.10      | 100.00      | 99.95         |
| Success regression    | 0.00        | 0.00       | 100.00      | 0.30          |
| Compliance regression | 0.00        | 0.00       | 100.00      | 1.95          |
| Joint gain            | 100.00      | 100.00     | 100.00      | 100.00        |
| Weak gain             | 3.50        | 0.05       | 0.15        | 54.65         |
| More ties, null       | 0.75        | 0.00       | 0.05        | 31.00         |
| More ties, cheaper    | 93.75       | 51.30      | 100.00      | 100.00        |

## J RELATED DECISION TARGETS AND THE COST OF DISJOINT PAIRING

**Stage reweighting and raw component requirements differ.** The priority-standardized net benefit of McCoy et al. (2026) combines stage-conditional effects using prespecified weights. Consider the same success-first, success-eligible cost rule as in Section 2. Let  $B$  always succeed at cost 2; let  $A$  fail with probability  $\epsilon = 0.1$  and otherwise succeed at cost 1. Success is the first tier. The cost tier is reached exactly when  $A$  succeeds, with probability  $1 - \epsilon = 0.9$ , and every reached comparison favors  $A$ . Thus the stage-conditional effects are  $(-0.1, 1)$ , ordinary net benefit is  $-0.1 + 0.9 = 0.8$ , and priority-standardized net benefit under weights  $(0.8, 0.2)$  is  $-0.08 + 0.2 = 0.12$ . Both composite effects are positive, while the raw success contrast  $-0.1$  fails a margin of 0.03. All stage reach probabilities are positive, and costs are decisive under the 5% tolerance. This algebra illustrates different population requirements. It does not show a miscalibrated test, compare operating characteristics, or claim novelty for the lower-tier-dominance concern.

**A standard projection identity at the same execution budget.** An efficient all-pairs estimator can reduce variance by reusing complete arm records. The following identity quantifies the fixed-prefix price of discarding those comparisons. It is the usual two-sample projection decomposition, not a new U-statistic result. It applies to the independent stationary arm law used by our primary score simulation; adaptive, nonexchangeable, or task-coupled records require a separate argument.

Let  $A_1, \dots, A_N \stackrel{\text{iid}}{\sim} F_A$  and  $B_1, \dots, B_N \stackrel{\text{iid}}{\sim} F_B$ , with independent arms. Both procedures use  $2N$  executed episodes:  $N$  records per arm. For a square-integrable comparison  $h$ , put

$$\theta = \mathbb{E}h(A, B), \quad a(A) = \mathbb{E}\{h(A, B) \mid A\} - \theta, \quad b(B) = \mathbb{E}\{h(A, B) \mid B\} - \theta,$$

and  $r(A, B) = h(A, B) - \theta - a(A) - b(B)$ . Define  $\sigma_A^2 = \mathbb{E}a^2$ ,  $\sigma_B^2 = \mathbb{E}b^2$ , and  $\sigma_R^2 = \mathbb{E}r^2$ . Consider

$$D_N = \frac{1}{N} \sum_{i=1}^N h(A_i, B_i), \quad U_N = \frac{1}{N^2} \sum_{i,j=1}^N h(A_i, B_j).$$

**Proposition 20** (Exact fixed-prefix variance comparison). *Under the independent-arm law above,*

$$\text{Var}(D_N) = \frac{\sigma_A^2 + \sigma_B^2 + \sigma_R^2}{N}, \quad \text{Var}(U_N) = \frac{\sigma_A^2 + \sigma_B^2}{N} + \frac{\sigma_R^2}{N^2}.$$

Consequently,  $\text{Var}(D_N) - \text{Var}(U_N) = (N - 1)\sigma_R^2/N^2 \geq 0$ .

*Proof.* Independence gives  $\mathbb{E}a = \mathbb{E}b = 0$  and  $\mathbb{E}(r \mid A) = \mathbb{E}(r \mid B) = 0$ . Thus  $a$ ,  $b$ , and  $r$  are orthogonal in  $L^2$ , so  $\text{Var}(h) = \sigma_A^2 + \sigma_B^2 + \sigma_R^2$ . The  $N$  disjoint scores are independent, proving the first formula. For the all-pairs statistic,

$$U_N - \theta = \frac{1}{N} \sum_i a(A_i) + \frac{1}{N} \sum_j b(B_j) + \frac{1}{N^2} \sum_{i,j} r(A_i, B_j).$$

The three displayed terms are pairwise uncorrelated. Two distinct residual terms are independent when they share no index. When they share only an  $A$  index, condition on that  $A$  record and use independent  $B$  records and  $\mathbb{E}(r \mid A) = 0$ ; their covariance is again zero. The same argument applies to a shared  $B$  index. Only the  $N^2$  identical-index residual variances remain, giving  $N^2\sigma_R^2/N^4 = \sigma_R^2/N^2$ . Adding the projection variances proves the second formula and the stated difference.  $\square$

If  $\sigma_A^2 + \sigma_B^2 > 0$ , the variance ratio is

$$\frac{\text{Var}(D_N)}{\text{Var}(U_N)} = \frac{\sigma_A^2 + \sigma_B^2 + \sigma_R^2}{\sigma_A^2 + \sigma_B^2 + \sigma_R^2/N}.$$

Its large- $N$  limit depends on the score law. If the first-order projections vanish but  $\sigma_R^2 > 0$ , the ratio is  $N$ ; if all three variances vanish, both estimators are constant and their ratio is undefined. At  $N = 1$  the estimators coincide.

For a component kernel  $h_g(A, B) = g(A) - g(B)$ , the residual is exactly zero, and  $D_N = U_N = \bar{g}_A - \bar{g}_B$  for every realized sample. This applies to the raw success and compliance gates in our

simulation. Reusing all cross-arm comparisons can improve the hierarchical score’s point-estimation variance but provides no additional component-mean observations on these same records. Different confidence boundaries may still have different behavior.

These fixed-prefix identities do not compare anytime error control, conjunction stopping time, delayed observation policies, or computation. All-pairs scores are dependent and cannot be inserted as  $N^2$  independent observations into the disjoint-pair betting process. The sequential U-statistic literature (Zhang & Wu, 2024; Bergemann & Hanson, 2026; Cai et al., 2026) addresses that dependence under its respective assumptions. No empirical superiority claim over those procedures follows from this identity or from our current reference experiments.